



Poet-Weaver: Reflecting on Communication Failure in Personal Relationships With Stylized AI-Generated Conversation Digests

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Interpersonal communication often involves navigating social challenges like managing expectations and conveying intentions. In practical contexts like productivity and navigation of social boundaries, agent- and AI-mediated communication (AIMC) has served as an effective social intermediary. Communication within personal relationships, especially between intercultural friends from different backgrounds, can also face significant challenges, such as miscommunication and expressive suppression. However, AIMC remains underutilized in the context of established personal relationships due to ethical concerns about agency and authenticity. We propose designing openly-interpretable AIMC output as an augmented context cue to reduce its active social involvement, balancing AI support with preservation of relational agency. We developed and evaluated Poet-Weaver, a Discord text chat plugin that presents AI-generated insights on user conversations in a stylized, openly-interpretable way to encourage reflection on communication challenges. We conducted a mixed-methods study with 30 intercultural friend pairs to assess Poet-Weaver's impact. Findings showed Poet-Weaver effectively helped participants address communication failures, although AIMC still influenced users' behavior even with openly-interpretable output. We recommend future uses of AIMC that support transformative, positive relationship changes while preserving individual responsibility and identity within relationships.

CCS Concepts: • **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Collaborative and social computing systems and tools**.

Additional Key Words and Phrases: AI-mediated communication, agency, interpersonal relationships, communication failure, social pragmatics, cross-cultural interaction

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1 Introduction

Many challenges and conflicts in interpersonal communication are intrinsically social, relating to differences in how people communicate and understand social situations [1, 6]. AI-mediated

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communication (AIMC), and in particular the social and language abilities of generative AI systems, have been shown to extensively support challenges in interpersonal communication by guiding users' navigation of interactions (e.g., suggesting what to say) or acting on behalf of them [39]. The unique ability of AI to function as a social actor with agency enhances its social impact compared to other types of technologies [40, 110], and explorations of AIMC so far have primarily been applied to practical contexts, such as managing interactions and boundaries in transactional, productivity-oriented, or educational settings [30, 34, 39, 69, 93].

However, interpersonal communication in the context of maintaining personal relationships can also suffer from significant challenges. Friendships from different social contexts (intergroup, or intercultural friends) struggle with achieving effective **metacommunication**: mutual understanding of how to interpret one another's communicated intentions [31]. This is fundamental for developing close personal relationships [26, 48], but intergroup friends are prone to misunderstandings that result from poor metacommunication and result in friendship dissolution [102]. Intergroup friends tend to transfer social norms across mismatching contexts, which generate gaps in understanding and cause **communication failure** [14, 21, 32, 97, 106, 108] during interaction between friends. For example, an intended compliment can be interpreted as an insult (e.g., "Wow, you really went all out" could signify praise, or imply excessiveness) [43]. There are significant barriers to resolving these issues and establishing metacommunication due to intergroup anxiety and expressive suppression that stem from negative social expectations [23, 42, 53, 102, 105].

This inherently social challenge (anxiety which prevents the initiation of interactions to address communication failure) [23, 106] presents a salient opportunity for support from AI, which can act as a social third-party to redirect social pressure and diminish this barrier by providing more explicit opportunities for individuals to initiate discussion [40]. However, robust applications of AIMC to support personal relationships have been limited [16, 27, 30], due to ethical concerns about overreliance on AI influencing the authenticity of personal interactions [28, 39]. Compared to more pragmatic contexts, AI impact on existent personal relationships can have enduring effects on human quality of life [48] by affecting critical issues like agency in social connection and relational identity (e.g., people's understanding of who they are to each other) [39, 41, 44].

To strike a balance between allowing social assistance from AIMC and preserving agency in personal interactions, we propose presenting the AI-generated perspective as an openly-interpretable social cue to be utilized during difficult-to-navigate interactions — this leverages the perceptual augmentation of context cues relevant to an interaction, a method of supporting interpersonal communication abundantly used in HCI (e.g., social translucence) [25, 35] which comparatively preserves user agency via minimal social engagement from the system. More concretely, we iterate upon and evaluate the following design concept:

The use of an AI-generated post-conversational summary as an openly-interpretable, non-instructive conversation prompt to support intercultural friends in jointly reflecting on prior conversations to resolve their communication failures.

Through an iterative design process, we designed a chat plugin, Poet-Weaver, which implements AI-generated conversational summaries **stylized** as a *tanka* (five-line poem) [84] to encourage open interpretation as a form of support for intercultural friends to resolve communication failure and engage in explicit metacommunication through joint post-conversation reflection. Poet-Weaver is **context-aware** by integrating user-marked conversation context from moments of communication failure into the AI-generated poem to provide friends with a third-party *bridge* for initiating challenging discussion about the failure. The goal of Poet-Weaver was to improve the quality of

metacommunication in intercultural friendships by supporting their ability to address communication failures they experienced. To assess these qualities, we aimed to answer the following research questions when evaluating Poet-Weaver:

RQ1 - How do stylized AI-generated conversational summaries and their context-awareness affect the **process** of establishing metacommunication (addressing communication failure) during collaborative reflection on a past conversation between intercultural friends?

RQ2 - How do stylized AI-generated conversational summaries and their context-awareness affect general **quality** of metacommunication and friendship between intercultural friends?

We then recruited 30 pairs of intergroup friends in a one-week mixed-methods, controlled field study to evaluate how well Poet-Weaver supported metacommunication and resolution of communication failures in a quasi-natural setting. To disentangle the effects of AIMC on intercultural friends as an unexplored population from the impact of context-awareness in Poet-Weaver, our study assessed three conditions (see [Section 3.3](#)): reflection without Poet-Weaver, reflection with context-unaware Poet-Weaver, and reflection with context-aware Poet-Weaver.

Our findings revealed complex and mixed effects of using AI-generated content support as AIMC in personal communication contexts. The mixed-methods analysis showed no significant differences across quantitative measures of metacommunication and friendship quality, but our qualitative findings revealed multi-dimensional effects of Poet-Weaver's context-awareness and presence. The context-awareness of stylized AI-generated summaries actively supported participants' discussion process to address communication failures and establish concrete metacommunicational strategies for future interaction. However, participants who reflected on their conversations without any AI-generated content support more strongly acknowledged their friends' behavior when judging the state of their friendship, reflecting a heightened sense of relational efficacy compared to those who used Poet-Weaver. Despite the minimal engagement and open-ended nature of the generated content, AI presence still influenced participants' sense of relational agency and freedom of expression during the interaction. This work provides novel empirical insights into the impact of minimal-engagement AIMC in a high-stakes personal relationship context and outlines future implications for AIMC's role in personal relationships.

2 Related Work

The following sections provide motivating background for our work on AI-mediated support as an openly-interpretable context cue to resolve difficulties in personal relationship communication. We describe current research on technology systems to support interpersonal communication and studies of communication challenges in personal relationships.

2.1 Augmentation and Mediation of Interpersonal Communication

2.1.1 Perceptually-Augmented Communication. Early explorations of enhancing interpersonal communication in the area of remote collaboration emphasized ways of augmenting context features that were relevant to the interaction at hand [35]. For example, Kuzuoka and Greenberg [56] used "digital surrogates" to represent physical actions by remote collaborators as context cues during communication. Other ways to improve the efficacy of interpersonal communication in a remote context have included multi-perspective video support to provide spatial cues for interaction [55], or visual captions to enhance comprehension of dynamic conversations during video conferences [61].

Systems have been designed to support communication in personal relationships, emotional expression, or accessible social interactions by enhancing contextual cues or provide new experiences and tools for users. In personal contexts, some of these systems provide contextual cues about daily activity [37, 57], while others provide context cues about emotional state [60] or use visualization to amplify emotional cues during text communication [3, 51]. To support physically-accessible communication, other systems enhance context cues in different modalities, such as speech bubble visualizations for hard-of-hearing users [75] or translating visual cues in video calls to audio cues for blind or low-vision individuals [90]. Augmenting the perception of contextual cues through technology has also been used to support cross-language communication, such as by visualizing conversation speed to increase social awareness [24] or visualizing gestures to encourage their use during multilingual conversation [58].

Perceptual augmentations of interpersonal communication support users in a manner that preserves agency – by providing cues, tools, or information to help them perceive, judge, and act more effectively during interaction.

2.1.2 AI Agent-Mediated Communication. In contrast, AIMC or agent-mediated interpersonal communication systems are designed to provide more socially-involved solutions to challenges during communication. The use of robots in moderating interpersonal communication for teams and children has been extensively studied [47, 86, 89], and AI-mediated communication has been prized for its utility as a “moral crumple zone” during challenging interpersonal interactions, which reduces the responsibility assigned to other humans when communication does not turn out as desired [40]. Generative AI (e.g., ChatGPT) as a more socially-complex and capable entity, tends to be employed in an instructional or intervening manner to support communication in more challenging interpersonal contexts, such as in online learning environments with unacquainted others [69] or to normalize conversations about consent and sexual boundaries [30]. Recent explorations of generative AI have demonstrated its potential for improving more affective and socially connective elements of interpersonal communication: Suh et al. [93]’s study on the use of generative AI during collaborative music composition revealed that it helped users develop rapport, while other systems such as HAILEY [88] increased conversational empathy in unacquainted peers.

However, the role of generative AI and AIMC in more personal and intimate communication contexts has been nuanced and controversial. Although social conversation partners evaluate each other more positively when supported by AI-generated suggestions, they develop negative evaluations about people believed to use generative AI [41]. Jakesch et al. [45] similarly found mistrust in personal descriptions suspected to be created by AI. AIMC systems designed to facilitate emotionally intimate conversations also found that users engaged in less emotionally-meaningful conversation when using AI [16]. Reviews and inquiries into AIMC have reflected on the potential impact it can have on maintaining meaningful relationships (e.g., perceived effort) [39] and authentic relationship interactions [28]. In Fu et al. [28]’s 2024 study on perceptions of AIMC, participants reflected a salient concern towards overreliance on AI especially in personal relationships; they were afraid that it would detract from the sense of agency and meaning in their relationships. Projects prior to the 2022 generative AI boom [8] have reflected this respect for interactional agency, especially in the context of personal relationships. For example, Kim et al. [52]’s 2019 “Love in Lyrics” employed AI-based recommendation as an inspiration for romantic partners’ text messaging while preserving users’ agency and control over their intended expressions.

The contrasts of user agency in perceptual augmentation and AI-mediated communication offer an opportunity to design systems that can support challenging interpersonal interactions without instructing or acting on behalf of the user. Our goal is to create a system that can leverage AI’s ability to support such social interactions as a third party mediating role, while still preserving

both agency and authenticity in personal interactions. Rather than bypassing users' judgment and behavior, our designed system Poet-Weaver stylizes AI output as an open-to-interpretation conversation prompt. It functions as a perceptual augmentation tool, enabling users to harness AI as a resource while still determining their own actions.

2.2 Mutual Understanding and Social Context in Personal Relationships

Due to their self-motivated nature, personal relationships are sensitive to dynamics in mutual understanding, which affect people's ability to communicate, interact, and socially connect with one another [19, 48]. Individuals from diverse social contexts, such as different cultures or personal backgrounds, often face significant communication challenges in personal relationships [23, 97]. These struggles, stemming from normative mismatches, can be difficult for dyads to overcome on their own and risk relationship dissolution [102].

2.2.1 Metacommunication and Development of Personal Relationships. **Metacommunication** is a multidimensional construct that relates to people's level of shared understanding regarding how they communicate with each other. Due to fragmented approaches towards studying metacommunication, this term can refer to either the quality of shared understanding about communication style or intention, or the process of successfully establishing that shared understanding [5]. For example, good quality metacommunication between two individuals can lead to correct interpretations of intended praise, sarcasm, or criticism from one another. Individuals can also engage in a metacommunication process to achieve better quality of shared understanding; this process can be explicit by exchanging clarifying statements of *semantic reassurance* [43], or implicit in the form of developing Polanyi's [77] *tacit knowledge* intuitively through embodied experience.

The quality of metacommunication has a significant impact on the fate of personal relationships. Because receivers' interpreted meaning of conveyed messages heavily depends on the context of how well they understand the sender and their behavior, there is a high level of variability in how interactions and relationship trajectories turn out [26]. Individuals with higher similarities in emotional processing experience more satisfaction in relationship interactions [50], while misunderstandings can harm personal relationships by changing the perceived nature of previous interactions and impacting future interactions [49]. Metacommunication quality also impacts the ability of individuals to develop their personal relationships by building rapport. Politeness theory emphasizes how people act to reduce threat to face in social interactions; when combined with poor levels of metacommunication, this behavior reduces individuals' comfort in acting honestly around each other [92].

2.2.2 Intergroup Issues in Metacommunication. Unsurprisingly, individuals from different cultures (defined by Thomas [97] as diverse geographical or personal backgrounds) are prone to experiencing challenges in metacommunication due to differences in social pragmatics. *Social pragmatics* represent the social context-dependency of language use during communication, and cross-cultural individuals are prone to experiencing *pragmatic failure*, which contributes to communication failure [15, 32, 97]. Cross-cultural struggles in establishing metacommunication stem from *pragmatic transfer*, which is when individuals project the social rules and conventions from their original context into a different, intercultural communication context [14, 108]. This can severely impact people's ability to achieve mutual understanding and de-escalate conflicts [32], express refusal [14, 42, 81, 108], or properly interpret sarcasm [72].

As a result, intercultural (intergroup) friendships are prone to a series of struggles that obstruct friendship development and maintenance by not only complicating the level of mutual understanding in proper metacommunication, but also preventing individuals from achieving it. Due to the awareness of differences in social knowledge, friends from different social groups or cultures

experience anxiety due to implicit prejudices about the other party and their expectations of misunderstanding or negative experiences [21, 23, 106]. Building common ground between cross-group friends has been shown to reduce such anxiety and negative expectations [74, 107], but resolving differences in understanding to achieve such common ground can be difficult for this group due to expressive suppression in cross-cultural communication. To avoid conflict when negative social experiences are anticipated, intercultural individuals engage in *expressive suppression*: where they suppress expressing or discussing negative emotions resulting from interaction [70, 85, 94, 105]. This behavior not only impedes metacommunication development and resolution of misunderstandings, but also stifles positivity in friendships [70] and inhibits friendship development [12].

Our work attempts to design a system which can support individuals to resolve communication failures and develop their quality of metacommunication by leveraging the third party advantage of AI as a “crumple zone” in this high-anxiety, high-suppression context where intercultural friends are unlikely to initiate such communication on their own.

3 Methods

We engaged in an iterative design process (Section 3.1) to refine our proposed idea to “provide openly-interpretable AI-generated conversational summaries as a context cue.” We conducted an initial interview probe (Section 3.1.1) of intercultural friends’ experiences with communication failure (defined in 2.2.2), which identified needs to be filled by the system, and a pilot study (Appendix A) to evaluate the stylization of AI-generated output for its open-endedness and fulfillment of user-identified needs (Section 3.1.2).

This resulted in the Poet-Weaver system, which generates five-line *tanka* poems describing conversations between friends (Section 3.2). We then conducted a mixed-methods, controlled field study (Section 3.3) to evaluate the impact of key system features on intercultural friends’ communication.

3.1 System Design and Implementation

In response to concerns about the over-engagement of AI in personal communication contexts [28], we explore the design of **AI-generated output as an openly-interpretable conversation prompt** for users to personally discuss on their own, as a more agency-forward way to alleviate the difficulty of jointly reflecting on communication failure which is experienced by intercultural friends (see Section 2.2.2). We posit that this type of *indirect*, openly-interpretable support is necessary due to the prior work’s demonstration of expressive suppression which prevents intercultural friends from speaking about communication failures at all. Yoshikawa’s *double-swing model* for framing intercultural communication [109] additionally reinforces this importance of open interpretation by explaining that fundamentally-different understandings between intercultural friends means there is no ‘ground truth’ of an interactional context that can be easily explained by one person to another. Therefore, resolving intercultural communication failures to attain mutual understanding requires both parties to actively and reciprocally “swing” between one another’s cultural perspectives, which is a daunting task. AI-generated material can serve as a mediating *bridge* to cushion the fear of ‘falling’ when engaging in such significant “swinging,” serving its role as a crumple zone [40].

This “inspiration-oriented” and openly-interpretable approach aligns with prior implementations of AI that prioritize user agency in expressive and emotional contexts [52, 103], in contrast to other more common directed or highly-engaged uses of AI to mediate communication (see Section 2.1.2). Openly-interpretable AI-generated content refrains from providing an answer to the user’s inquiry, and does not provide instruction or conduct any behavior on behalf of the user.

The following sections will detail our iterative process for creating and evaluating such open-ended AI-generated conversation prompts. We performed pilot-testing ([Appendix A](#)) on two versions of the generative AI prompt with five pairs of intercultural friends before evaluating the full system ([Section 3.2](#)) in a mixed-methods field study ([Section 3.3](#)).

3.1.1 Determining Core System Features. We conducted 30-45 minute interviews with six individuals in intercultural friendships to probe their detailed experiences of communication failure (when friends fail to convey or understand a message as intended, see [Section 2.2.2](#)). We then derived key system needs based on interviewees' commonly-described concerns.

Our interview questions probed the specific circumstances of intercultural communication failures, building upon prior work ([Section 2.2.2](#)) and informed by the lack of inquiry into specific mechanisms of how expressive suppression manifests from intercultural anxiety. In order for our implementation of AI mediation to address immediate needs in this context, we aimed to understand the relevant aspects and granular concerns of this interactional issue. We asked interviewees about situations during their friendship where they felt misunderstood, offended, or had difficulty communicating something to each other. We then elaborated by asking whether they were able to address the difference in understanding, or whether it was difficult to do so. If it was, we asked them to explain the detailed situation of why it was hard to resolve their differences in understanding. We also asked about the impact of these issues on their overall friendship with this person.

This initial interview probe collected intercultural friends' grounded experiences with communication failure and their social concerns which prevented addressing those failures through a metacommunicative process. Some interviewees described situations where they were unsure whether their friend intended to say something offensive to them, but refrained from discussing it due to fear of "*making a big deal*" by going off topic. Although they chose to move on conversation-wise, these experiences had lasting negative effects on their emotional state during and after the interactions. One person recalled a time when their friend made a remark about their occupation that they perceived as condescending, but they ignored it and continued the discussion despite lingering negative emotions. Other interviewees described moments when their friends seemed unexpectedly upset at something they had said, which made them feel as if their original intention failed to come across. But nearly all interviewees stayed silent in these situations and attempted to mask their negative internal reactions out of fearing awkwardness or exacerbating the conflict.

Based on interviewees' experiences and concerns, we identified two key needs of support for intercultural friends to address communication failure:

- (1) **indirectness** of conversational support or reference to communication failure
- (2) **subjective context** of communication failure

Indirectness was a key requirement we identified, as interviewees' expressed extreme aversion towards the communication failures they experienced being brought up or shown directly to each other. But in order for the communication failure to be addressed, they also expressed a need for something other than themselves to present the issue so they could talk about it. Interviewees felt that the act of bringing up an issue itself had a strong directionality and would have adverse consequences if misunderstood. This aligns with prior work showing conflicting strategies across cultures of how people attempt to preserve social harmony during perceived conflict [71]. Intercultural individuals are largely unaware of the conflicting interpretations they have in these contexts, which leads to snowballing of mismatches in intended management strategies and results in judging each other as uncooperative during attempted resolution of differences [101].

Subjective context of communication failure (informed by the user) was also necessary to support actual referencing and addressing of communication failure, since interviewees reported that they actively masked their experiences of communication failure due to fear of causing

conflict. This made it nearly impossible for anyone other than themselves (human or AI) to detect the occurrence of communication failures. The absence of outwardly-perceivable indicators of intercultural communication failure corresponds with the implications of *expressive suppression* (Section 2.2.2), which implies that individuals will suppress any indication of the experienced failure. The importance of representing subjective context to assist the process of resolving communication failure also reflects key principles of the *double-swing model* [109], which asserts the need to ‘swing’ between the subjective truths experienced by each culturally-disparate individual.

The following section 3.1.2 describes how we implemented this requirement by allowing each individual user to **privately mark chat messages** associated with a communication failure, which would then be integrated into the AI-generated digest of the conversation.

3.1.2 Creating the AI-generated Conversation Prompt. We explored the design of **AI-generated output as an openly-interpretable conversation prompt** as a very minimal form of support to users’ joint reflection on their communication failure. We chose to implement collaborative joint reflection as the mode of support due to the aforementioned issues, such as interviewees’ aversion towards coming off as adversarial or exacerbating the original misunderstanding, as well as known benefits of social reflection for motivating daunting behavioral changes [76]. In designing for **open-interpretability**, **indirectness**, and **subjective context**-integration, we chose to evaluate two initial designs for the language style:

- A **plain-language** summary of the conversation, formatted into bullet points to better facilitate comprehension
- A **poem**-format summary of the conversation in an aesthetically-stylized manner

We evaluated these two stylistic options to support creative, open interpretation within an intercultural context, the ability to inject indirect and subjectively-defined conversational references, and to mitigate the known mental influence that AI-generated conversational prose has over users [29]. In addition to a straightforward plain-language summary which also included indirect references to marked subjective context, we chose to evaluate a poetic format due to its linguistic properties that depart from the normal flow of speech. This stylistic technique, coined in 1932 as *foregrounding* by Czech literary theorist Jan Mukařovský [68], refers to purposefully making parts of the text stand out from its surrounding context. This quality is absent in plain language.

Foregrounding in poetry deautomatizes the perception of information through techniques such as deviation from linguistic norms, parallelism, metaphors, and leveraging the *polysemy* (multiple meanings) of words. Foregrounding promotes *defamiliarization* (reverberating Bell et al. [7]’s “Making by making strange”), which prompts readers to reconsider different perspectives about the presented text, feel stronger affect, and think more deeply [67]. In addition to the metaphoric and polysemic features that promote our goals of **open-interpretability** and **indirectness**, *foregrounding*’s provocation of schema-refreshment (conventional ways of thinking about the world being disturbed and accordingly refreshed) [36, 87] can support the goal of helping intercultural friends reach acceptance of multiple **subjective contexts** [99] and reduce concerns about overreliance on AI-generated conversational prose [2].

Both types of conversation prompts were developed using GPT-4o to **indirectly** integrate **subjective context** of communication failure. After probing individual experiences of communication failure with friends through interview, we decided to implement this functionality by enabling users to **‘privately mark chat messages’** associated with a communication failure, such as a message containing content that they had doubts about but could not address directly. During AI-generation of the conversation summaries, the prompt would then instruct GPT-4o to contain indirect references to those specific marked moments.

After initial experimentation with the length and content of AI-generated summaries for 15-minute text conversations (the average length of conversations assessed in relationship research [63]), the average length of a concise plain-language summary from GPT-4o was around five bullet-pointed sentences generated by the AI. Therefore, we decided to use the five-line *tanka* [84] as the format for the AI-generated poem (see Figure 1). Although the plain-language summary is visibly longer than the poem, this was the most concise output that we could get GPT-4o to generate without losing core elements of personal conversations. The verbosity of dialogic responses from LLMs is a well-known problem [10, 83].

We also used DALL-E-3 to generate imagery representing the content of each summary to pair with the text (Figure 1), in order to support intercultural compatibility during interpretation of the text [33, 98] and encourage open-ended thinking during the reflective discussion process [17, 59]. We conducted a **pilot study** to compare these two language styles before integrating the final conversation prompt into the system (see Appendix A for details on the pilot study and Appendix B for the final prompts used in the system).

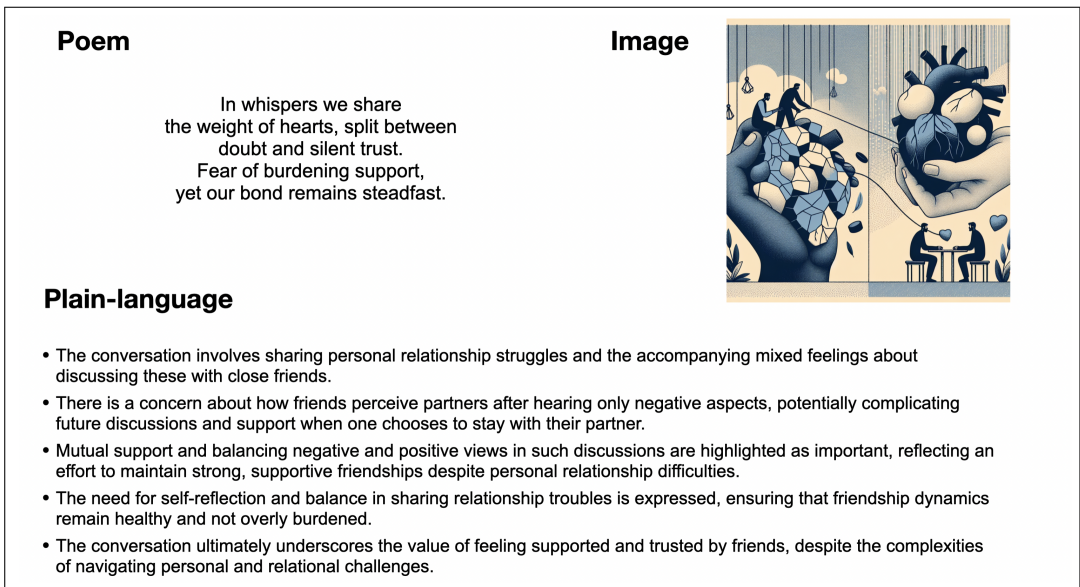


Fig. 1. **Examples of generated summaries for each style and corresponding imagery.** Summaries had to fulfill the condition of being indirect when referencing parts of users' conversations.

3.2 Poet-Weaver System

The Poet-Weaver system (see Figure 2) is a Discord¹ plugin which generates a poem and paired image (see Appendix B for the final prompts) based on a prior conversation between users to serve as a discussion prompt for reflecting on their conversation. The system functions by facilitating “*conversation-reflection sessions*” through a **5-step** interaction process:

- (1) Users enter the command `convo_start` into the Discord chat when they want to begin a *conversation-reflection session*.

¹<https://discord.com>

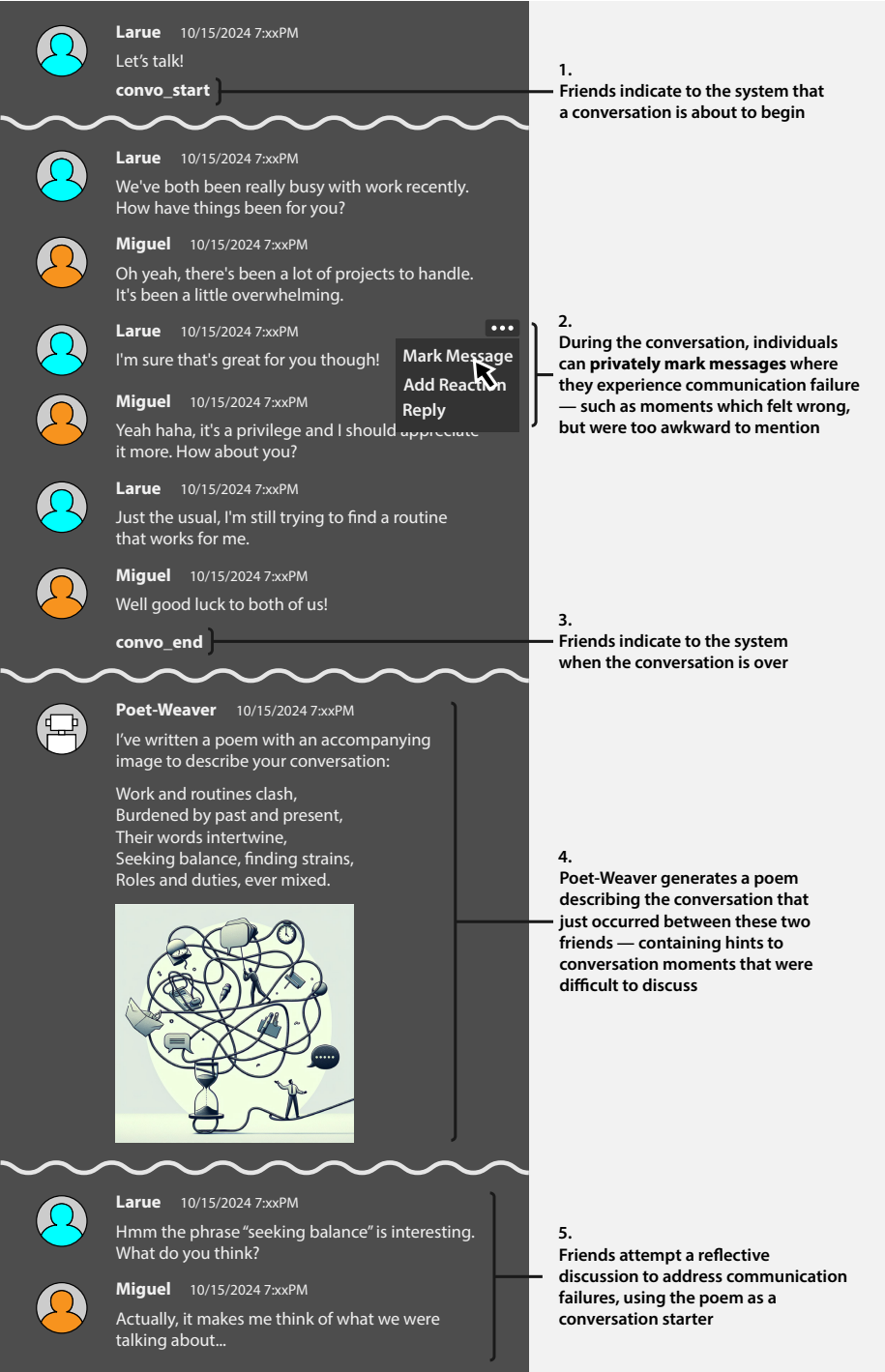


Fig. 2. Poet-Weaver interaction flow for a conversation-reflection session.

- (2) During a **conversation**, users can **privately mark chat messages** where they perceive communication failure, such as messages which felt wrong and seemed to potentially reflect a misunderstanding, but which users could not bring themselves to address (see 2.2.2).
- (3) Users enter the command `convo_end` into the Discord chat when they want to end the conversation portion of the session.
- (4) Poet-Weaver writes a poem with an accompanying image based on the conversation denoted by the start and end commands. This is shown to both users along with a message to reflect on their conversation. The contents of the poem and image contain **indirectly-presented** integrated elements of both user's marked messages from the prior conversation.
- (5) After seeing the generated poem, both users initiate a **reflection** session where they attempt to discuss and reflect on the conversation they just had, with the goal of addressing any communication failures experienced (a form of establishing mutual understanding as a part of metacommunication).

3.3 System Evaluation Study

3.3.1 Study Design. We conducted a mixed-methods controlled field study with intercultural friends to evaluate the effects of AI-generated support, as well as the effects of AI-integrated context of communication failure, on addressing the communication failures they experienced. Because there have been no studies on the impact of AIMC on this social context, we wanted to distinguish effects of AI involvement from the specific effects of system components relevant to addressing communication failures and establishing metacommunication.

Therefore, we decided to evaluate three conditions in a between-subjects design:

- **Reflection without Poet-Weaver (RWP):** After their conversation, participants receive a simple prompt in the chat system indicating that they can start to “reflect on the conversation you just had.” There is no additional support or information provided.
- **No-context Poet-Weaver (NCP):** After their conversation, participants receive a message from Poet-Weaver that presents a poem and image created based on their conversation, and a prompt to “reflect on the conversation you just had.”
- **Context-aware Poet-Weaver (CAP):** After their conversation, participants receive a message from Poet-Weaver that presents a poem and image created based on their conversation, and a prompt to “reflect on the conversation you just had.” The poem and image here integrate subjective context of communication failures which were privately marked during the conversation (see Section 3.1.2).

To approximate participants' conversational experiences in the wild, we evaluated the system in a field setting. We assigned each pair of participating friends a private Discord channel which they would use for one week instead of their usual text messaging application to communicate. Aside from their regular text messaging practices, each pair of friends engaged in two measured **conversation-reflection sessions** (see Section 3.2) during this one week period. They could select the time to have these conversation-reflection sessions at their own convenience. To control our measures for the study, we limited conversations and reflections to 15 minutes each (30 minutes for each conversation-reflection session).

Because communication failures happen serendipitously, we strived to facilitate a conversation environment in which they might occur by asking participants during onboarding to select topics for their conversation-reflection sessions that aligned with what they talked about on a daily basis but had led to communication failures in the past or seemed likely to. Examples of topics picked

by participants included social experiences at work or school, managing interactions with mutual friends or colleagues, and interactions with family members or significant others.

All private Discord channels assigned to participants contained a plugin that enabled us to facilitate the conditions of the study. The CAP condition included all system features, such as responding to start and end of measured conversation-reflection sessions, AI-generation of the poem and image, and the integration of users' privately marked messages into the AI-generated content. The NCP condition included all system features except for private message marking and integrating context into generated content. The message marking feature was not introduced or enabled in this condition. The RWP condition included a plugin which only responded to the start and end of the conversation portion (Steps 1 and 3 in [Section 3.2](#)) with a fixed message indicating that the participants can start to reflect on their conversation.

This study was approved by the Institutional Review Board at the author's institution.

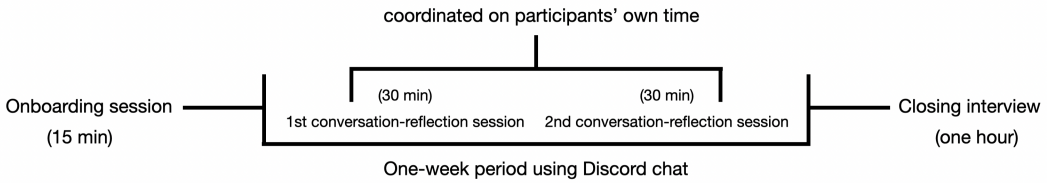


Fig. 3. **Procedure of mixed-methods controlled field study for one participant pair.** Each pair of participating friends engaged in two conversation-reflection sessions (3.2) over the course of the study.

3.3.2 Procedure. The study procedure mainly consisted of three phases: (1) an onboarding meeting to introduce the system and select two conversation topics, (2) a one-week period during which participants used the system and engaged in two conversation-reflection sessions (one for each topic), and (3) a closing interview to share their thoughts on the experience.

We reached out to eligible pairs of friends over email and asked them to digitally sign a consent form, fill out an initial demographic survey, and initial responses to the McGill Friendship Questionnaires (MFQ-RA, MFQ-FF) and the adapted Metacommunication in Supervision Questionnaire (MSQ) which we used to quantitatively measure friendship and metacommunication quality (see [Section 3.3.4](#)). We then scheduled a 15-minute onboarding meeting with each pair of friends. During the onboarding meeting, the experimenter facilitated participants' selection of two conversation topics, introduced how to use our plugin in Discord, and reviewed the instructions for participants' one-week use of the system before scheduling the closing interviews. Participants were allowed to communicate on Discord of their own accord and engage in the two 30-minute conversation-reflection sessions on their own time (see [Section 3.3.1](#)), after which they asynchronously filled out post-study responses to the MFQ-RA, MFQ-FF, and MSQ. Closing interviews with each participant lasted one hour. Participants were compensated with the equivalent of a 45 USD gift card upon completing the study.

3.3.3 Participants. We recruited 30 pairs of intercultural friends (60 participants, 10 pairs in each condition) through social media and cross-institutional mailing lists. None of them had previous knowledge of, nor had they participated in, our interview probe ([Section 3.1.1](#)) or pilot study evaluating the conversation prompts ([Appendix A](#)). Interested pairs of friends filled out a screener survey with their demographic information, cultural background, and perceived level of closeness using the Inclusion of Other in the Self scale (IOS) [4]. From this, we recruited friends with valid responses describing differences in cultural background based on Thomas [97]'s definition (see

Section 2.2.2) and IOS higher than one (out of seven) who were available to participate in the study. Our sample included 33 female, 25 male, and two non-binary participants. Participants were between 20 and 38 years of age ($M = 26.47$, $SD = 4.42$). The main type of cultural difference reported by participants was cross-national cultural differences (growing up in different countries), followed by differences in ethnicity, religion, socioeconomic status, and family background. Due to the large number of participants, see **Appendix C** for the full demographics table.

3.3.4 Measures and Analysis. To answer **RQ1** (effects on the process and experience of establishing metacommunication) and **RQ2** (effects on quality of metacommunication and friendship), we employed both quantitative and qualitative measures. Please see **Table 2** for a summary of how our findings connected to the RQs.

We conducted thematic analysis [11] on the 60 interviews conducted across the three conditions and cross-checked participant reports with the content of their conversation-reflection sessions, which resulted in the themes described in **Section 5**. The first author open-coded the interviews, focusing on identifying similarities and differences across the three conditions in terms of participants' process to establish metacommunication, the metacommunication established (communication failures addressed), and participants' feelings regarding their friendship. All co-authors then reviewed emerging subthemes across three discussion sessions before finalizing the themes.

For quantitative measures of friendship and metacommunication quality, we employed a condensed version of the McGill Friendship Questionnaire set for friendship affection (MFQ-RA, 9 items) and function (MFQ-FF, 10 items) [66] and adapted the language in Metacommunication in Supervision Questionnaire (MSQ, 8 items), an existing validated measure for frequency and willingness to engage in metacommunication [13], for a friendship context. The specific questionnaire items are provided in **Appendix D**. We used the Kruskal-Wallis test to analyze pre/post-study changes in these questionnaire items across all conditions.

4 Quantitative Results

Overall, our quantitative measures of friendship quality (MFQ-RA, MFQ-FF) and metacommunication (MSQ) did not show significant differences across conditions.

Responses in the MFQ-RA were divided into *Satisfaction* (4 items) and *Positive Feelings* (5 items), with before-after difference scores analyzed using the Kruskal-Wallis test. For *Satisfaction*, the median and interquartile ranges were as follows: RWP: 0.0 [-1.0, 1.0]; NCP: 0.0 [-1.0, 1.0]; and CAP: 0.0 [-0.25, 1.0]. For *Positive Feelings*, the median and interquartile ranges were as follows: RWP: 0.0 [-0.25, 1.0]; NCP: 0.0 [0.0, 1.0]; and CAP: [-1.0, 0.25]. The Kruskal-Wallis test indicated no significant differences across conditions for both *Satisfaction* ($p = 0.973$, $\eta^2 = -0.016$) and *Positive Feelings* ($p = 0.389$, $\eta^2 = 0.015$).

For the MFQ-FF, responses were divided into *Emotional Security* (3 items), *Intimacy* (3 items), and *Reliable Alliance* (4 items). For *Emotional Security*, the median and interquartile ranges were as follows: RWP: 0.0 [-1.0, 2.0]; NCP: 1.0 [0.0, 3.0]; CAP: 0.0 [-0.25, 2.0]. For *Intimacy*, the median and interquartile ranges were as follows: RWP: 1.5 [-0.25, 3.25]; NCP: 0.0 [-1.0, 3.25]; CAP: 0.0 [-1.0, 3.0]. For *Reliable Alliance*, the median and interquartile ranges were as follows: RWP: 0.0 [-1.25, 1.25]; NCP: 0.5 [-1.0, 3.0]; CAP: 0.0 [-2.0, 1.25]. The Kruskal-Wallis tests for each subscale revealed no significant effects of condition on *Emotional Security* ($p = 0.596$, $\eta^2 = 0.0006$), *Intimacy* ($p = 0.605$, $\eta^2 = 0.00009$), or *Reliable Alliance* ($p = 0.528$, $\eta^2 = 0.0047$).

Similarly, analysis of the MSQ, based on the sum of eight questionnaire items, revealed the following medians and interquartile ranges: RWP: 1.5 [-2.0, 6.0]; NCP: 3.0 [-2.0, 5.5]; CAP: 0.5 [-2.25, 5.25]. The Kruskal-Wallis test showed no significant difference across conditions ($p = 0.657$, $\eta^2 = -0.0027$).

While our quantitative measures did not show significant differences across conditions, our interview findings demonstrated qualitatively different impacts of each condition on participants' reflective interactions, uncovering effects not captured by the quantitative data.

5 Interview Findings

Our Findings identify salient themes of how the context-awareness and presence of AI-generated support impacted friends' ability to engage in metacommunicative reflection. [Section 5.1](#) shows how context-awareness-dependent features of the AI-generated poems in conditions CAP and NCP impacted the process to resolve communication failure (RQ1), as well as the strategic and emotional outcomes (RQ2) of this process. [Section 5.2](#) presents our observations of how the presence of AI-generated support impacted participants' own sense of friendship (RQ2), and [Section 5.3](#) highlights the double-edged effects of AI-generated content as a mediator during social reflection (RQ1). Please refer to [Table 2](#) for an explicit summary of connections to the RQs.

Participant quotations in this section are *italicized* and denoted with “double quotes,” while quotations cited by participants or lines from the poem are denoted inside ‘single quotes.’

5.1 The Impact of Context-Awareness in AI-Generated Poems on Cross-Cultural Friends' Ability to Reflect on Failure and Grow Closer

Our analysis identified clear trends in the impact of subjective context-awareness in the AI-generated poems on participants' success in resolving their communication failures (RQ2) and engagement in explicit metacommunication during the reflection process (RQ1). To make sense of these results across conditions, we coded participants' experiences for every communication-reflection session based on the following definitions:

- **“brought up/resolved”** - the person had a metacommunicational issue (communication failure) during their conversation and successfully addressed it during the reflection
- **“didn't bring it up/unsuccessful”** - the person had a metacommunicational issue and did not bring it up in their reflection, or tried to bring it up (e.g., indirectly) and was unsuccessful in directing the conversation towards addressing the issue
- **“no concern”** - the person had no metacommunicational issue in that conversation

	brought up / resolved	unsuccessful / did not bring up	conversations with perceived issues
Reflection without Poet-Weaver (RWP)	6	18	24
No-context Poet-Weaver (NCP)	8	17	25
Context-aware Poet-Weaver (CAP)	19	9	28

Table 1. **Resolved communication failures based on each participants' experience during reflection.** Counts are based on 40 conversation-reflections per condition (n=20), with two experiences per participant.

Based on the reported counts in [Table 1](#), participants using CAP experienced the highest number of resolved metacommunicational issues. Our thematic analysis reveals many participants using CAP brought up the communication failure as a metacommunicational issue (reflected in their marked message) by discussing specific parts of the generated poem. In contrast, most participants using NCP discussed general topics guided by the poem content instead of reflecting on any part of their prior conversation. These trends were influenced by divergent ways in which context-awareness of poem content (CAP and NCP) directed participants' reflective discussions.

Morning greetings shared, Dream tales, bridging thoughts on ties, Gender lens explored, Echoes of trust and tension, Friendship's thread through day and night.	Beneath smiles unshown, Masked feelings and silent thoughts, Conflicted hearts held, Policy vs. true words, Friendship's complex gentle sway.
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Fig. 4. **Real conversation poem digests generated by Poet-Weaver.** The no-context poem (NCP) is on the **left**, and the context-aware poem (CAP) is on the **right**. Although they do not have striking differences on the surface, participants actually involved in the conversations used as input tended to respond differently.

The differences in the two poems shown in [Figure 4](#) are not apparent to a third party. They both have five short lines and contain a similar amount of detail in the keywords used. However, the parties involved in the conversations used to generate these poems were sensitive to keywords that related to specific parts of their conversation. Hence, the context-awareness of the poem in relation to moments of communication failure made a difference to participants using CAP compared to NCP. Specifically, CAP poems (in multiple cases) allowed participants to re-address and discuss uncomfortable communication issues from their initial conversation, while most participants responded to NCP poems by keeping the discussion more general and positive.

Participant 1A (CAP) was able to use the poem in [Figure 4](#) (right) to bring up an unspoken concern from her initial conversation with P1B. They had been speaking about their relationships with other friends, and P1A felt uncomfortable when she noticed a difference in understanding between her own and her friend's attitudes towards others. She marked those messages from her friend, but was not able to express her feelings in the initial conversation. She reported that during the reflection, *"the poem made it easier to tell our true feelings. It gives us some hints that we can use to open the discussion."* She was inspired specifically by the phrase 'masked feelings' in the poem, which she felt captured the specific part of the conversation where she experienced the issue. It was easier to ask her friend, *"Do you think you are masking your feelings?"* which led them to discuss, acknowledge, and better-appreciate one another's differences from the initial conversation.

In contrast, P12A (NCP) was not able to use the poem in [Figure 4](#) (left) to resolve an unspoken concern from her conversation with P12B. She had felt that her friend's response to her discussion of gender-related experiences was superficial, but chose not to bring it up with him. Although the poem reflected the themes of their conversation, it did not allow her to discuss her concern during the reflection. Although *dreams* and *gender* were conversational themes, the poem did not motivate P12A to reflect on their conversation. As a result, P12A stated *"we leaned towards finding a happy conclusion for the conversation we just had. I think it's difficult to address challenges in our communication, because we like to look at things that work, but not things that are not understood."*

5.1.1 Context-aware poem effects: enlightening discussions and metacommunication strategies.

Enlightening discussions. The majority of CAP participants described how specific lines from the poems and poetic features such as polysemy, emotional expressivity, and neutral existence helped them to discuss and address the concerns they had marked in the initial conversation.

They highlighted how the **polysemy** of the poem's references to marked conversation parts allowed them to broach the issue without negative biases from either side. For example, P7A experienced and marked a communication failure while conversing with P7B about a mental health issue they both shared. P7B gave advice that P7A felt was dismissive and invalidating of his experience, and *"it wasn't what I wanted to hear at the moment. I just wanted support... But then we changed the topic to something more relatable."* Because the issue was brushed over in an effort to avoid conflict, P7A failed to address this through normal conversation. However:

“When we read the line of the poem, ‘Linked paths, their struggles unveiled’ it showed that we both went through tough times, this connection between us that was relatable in both senses. When we read that, it was like, oh, yeah. Both what you were saying, and what I was saying is relatable. So we started with that.”

By mentioning the relationship between the friends’ experiences, this line of the poem served as a bridge for P7A to discuss the uncomfortable perceived disconnect with P7B in their experiences and resolve their rift in understanding:

*“At first, it seemed like someone who’d never been through that saying, ‘You just have to be strong enough.’ But it turned out that he also shared the same experience. After I understood that, it made more sense why he said what he said, and I felt more supported. **Through the conversation, he broke down the prejudice that I had.**”*

Similarly, P14B described how the polysemic references to religion in their poem helped she and P14A resolve a misunderstanding that P14A had about P14B’s religious orientation which she discussed during the conversation:

“I was talking about feeling guilty for something connected to my religion. [My friend] thought she made me uncomfortable talking about that, but we did not clarify that in the initial conversation. Then there were some religious aspects in the poem. So she referred to it, and I told her that actually, I was aiming to tell a joke about it.”

In the conversation, P14A and P14B were discussing sexual topics when P14B mentioned that she grew up with a religious *taboo* against it — P14A misunderstood this as a traumatic orientation towards the topic and marked it as a failure for bringing it up to P14B. However, the taboo-related words “*silent*” and “*hushed*” in the subsequently-generated poem opened a natural opportunity for P14A to make a reference and follow with a “*By the way...*” statement to bring up and clarify the issue. Others echoed how the referential ‘bridging’ quality of the poem’s polysemic references to problematic conversation landmarks allowed them to approach talking about the implicit conflict more confidently. P28B described how:

*“It was more comfortable. Sometimes, mentioning [the issue] directly can be hard. **And the poem helped me to focus.** Like, ‘Guys, focus your issue here and talk about how you can deal with this, or how you can reflect on this.’”*

The ‘bridging’ quality of the poem supported participants’ ability to initiate: “*It gave me a shortcut to say what I wanted to talk about,*” (P7B) and “*helped me think about how to more clearly explain what I wanted to say.*” (P13A)

Several CAP participants also mentioned how the expressively ‘poetic’ nature of the poem brought an emotional and personal quality to their reflective discussion, creating space for them to open up more vulnerably to one another. P15A, who awkwardly felt pressured by P15B during a conversation about her personal life, mentioned that “*We felt like we tried to maximize the emotions in the poem and the images. The tone was more inspired, so the atmosphere was more positive.*” This helped them to address P15A’s perceptions and feelings about her boundaries without worrying as much about inadvertently appearing accusatory or hostile in the process. P14A also stated how “*Poetic language was used to describe a stigmatized topic [sex], and we really related to it. The image was very peaceful and gentle which set the tone for our discussion.*” In fact, she also used the content of the poem to crack a joke (the poem’s phrase “*touched openness*” inspiring her to say “*I am open for touching, jk*”), which set a more emotionally-open atmosphere for talking about potential miscommunications.

Others described how the neutral placement of the polysemic poem material into their shared conversation space did not just make it easier to ‘bridge’ to difficult topics, but also stimulated their

ability to think more critically when engaging in reflection. P28A mentioned how the abstract and open-ended nature of the poem helped him to think more openly about the communication failure he experienced, where he felt like P28B was being contradictory in complaining about his struggle with a workplace communication issue:

*“The poem helped us as we try to interpret things that we didn’t talk about earlier. **It’s something that we read and we either agree or disagree**, as it reflects the situation and help us to discuss further about the situation. Without the poem, we’d miss some moments so we’d likely ignore the situation and just move forward.”*

The phrases of their poem “*Nuances lost in the mail*” and “*Rushed tasks*” referenced the main breakdown in the conversation where P28A had assumed P28B used email in lieu of better communication options during a rushed deliverable. In revisiting the specific part of P28B’s narrative that he perceived to be contradictory, P28A realized through their discussion that his assumption about P28B’s options was not true. On the other hand, P26B described how the mere presence of the poem material as a ‘neutral party’ in the conversation “*lowered the barrier of just saying what I want to say*” in asking whether P26A was repressing her displeasure about something he shared:

*“It felt like there was some third entity being there which made it easier to speak freely. Before, if my friend was unhappy about our conversation I would be **more careful analyzing it. Here is the opposite - I am more open in analyzing it.**”*

Many participants reported feeling a sense of meaningful change to their friendships after the reflective discussion. For example, they mentioned feeling more “*supported*” (P7B, 28A), “*calmer*” (P1B, 28B), “*relieved*” (P26A), “*more empathy*” for their friend (P14B, 16A), and more “*trust*” (P28B). P28B was grateful for the “*change in understanding*” and P26B appreciated “*not feeling like anything I have to say is weighing on my mind anymore.*” Many of these feelings such as trust, support, and empathy are significant contributors to relationship quality.

Metacommunication strategies. Having gone through the process of addressing their communication failures, CAP participants came away with concrete metacommunicational strategies to enhance mutual understanding in their friendship. For example, P14B felt “*inspired to dedicate a bit more time to our friendship,*” after noticing her friend’s needs during their reflection. Most CAP participants expressed increased willingness to talk about things they did not agree on, and others went beyond by developing strategies to “*ask more clarifying questions*” (P5A), clearer ideas of how to discuss “*our priorities for plans together*” (P26A), and being more transparent “*that I’m still interested in the friendship and not just focused on myself*” during times of unavailability (P16A). P14A also mentioned how the “*reflection helped me understand the meaning beneath the words,*” a metacommunicative strategy that she learned for future interactions:

*“I think I sometimes don’t notice the conversation structure, which makes the dialogue ineffective. I only focus on the words that people mention, and just think about the words rather than reflect on the overall structure. **So instead of only focusing on words, I want to focus on the motivation of what someone said.**”*

However, a few CAP participants reported choosing not to address their communication failure in the reflection, even after identifying specific parts of the poem that they could use as a bridge to discuss their issues. They described reasons such as an inherent discomfort with talking about the issue (P13A), or feeling as if bringing it up would not make a difference (P19B).

But overall, qualities of the CAP poems such as pointedness of content, emotional quality, and presence in the conversation enabled many participants to address conversational issues during

their reflection. Participants came away with significant feelings toward the friendship and concrete metacommunicational strategies for future interaction.

5.1.2 No-context poem effects: Distracting, but positive for bonding. Most NCP conversations were characterized by avoiding conversation challenges and instead embracing opportunities for positive bonding. Participants reported that their conversations were shaped by the generic and imprecise nature of the NCP poems. P6A reported that *“the poem gave the gist of the conversation, but it was vague.”* The lack of compelling references in the poem caused her to *“just accept what [my friend] was saying.”* during the reflection. Participants recalled that the NCP poems felt *“positive”* (P12A) and *“friendly”* (P8A), but that it was *“basically the summary of what we talked about.”* (P3B). This distracted participants from reflecting during the reflection phase, instead prompting them to discuss the broader topics reflected in the poem. P29A explains:

“I think in the [reflection], I didn’t mention [the issue I had] because we started off with the poem. We started talking about our situation as a whole. So then I didn’t really think about the feeling I felt during the initial conversation.”

Instead of reflecting on their interaction and addressing communication issues, NCP participants ended up talking about broad issues such as culture and philosophy (P22AB), navigating family (P12AB) and academic situations (P10B), or the importance of sharing difficulties with others (P29B). Almost half (eight) of the NCP participants mentioned that *“the reflective discussion did not make any difference”* for their friendship. Some even emphasized that *“the poem... was difficult to use as a connection point”* (P10A). P22A felt that *“we would have been more into [the reflection] if the poem wasn’t provided. I think we would have brought up more specific points from the conversation.”*

Despite being unable to reflect on their specific communication failures, many NCP participants described reinforcing positive feelings and views through discussing the NCP poem. P29A mentioned that *“we wouldn’t have had as deep of a conversation without the poem,”* which enabled her to see more of P29B’s supportive side despite her discomfort with a perception of status difference (the original communication failure). P22A, who had felt that P22B was dismissive of her cultural experiences, was still able to discuss cultural experiences at a broader scale based on the poem. She mentioned that *“the poem can give a perspective that we both didn’t notice.”*

After the reflections, NCP participants often reported feeling positive about their friendship due to the process of affirming positive feelings and views with their friend. P3B mentioned how, *“It’s not something new, but talking about this again is nice because it makes you have a deeper connection with the person.”* P6B and P22A both mentioned feeling better after confirming in the reflection that *“we feel the same”* about some of the things they were talking about. Some of the positivity appeared to stem from the aesthetic nature of the NCP poem, which sometimes encouraged positive discussions around social issues. P29A mentioned a *“line like, ‘friendship’s strength carries us through.’ So we were like, oh this gets us and it’s like we are kind of teaming up.”* Similarly, P12A described an overall positivity conveyed through one of their poems:

“Our first conversation evolved into talking about how [friend] was feeling homesick. And the poem put it in a beautiful way explaining how you’re missing home in a different country. And the image also. The poem felt like a really wise, old man.”

NCP participants generally concluded upon some general positive effects for their future communication, such as more willingness to share hard things with each other, due to feeling closer (P29AB), less awkward (P8A), and more listened to (P29A).

Overall, NCP participants tended to have broader discussions during their reflective phase and attempted to keep things positive instead of addressing specific issues from their initial conversation. Participants reported being affected by the *‘generic’* nature of the poem, but many also felt a stronger

sense of positive affinity with their friend after the discussion, which led to more willingness towards future sharing and open communication.

5.2 The Degree of AI Intervention Relates to Friends' Perceived Relational Efficacy

Perceived **relational efficacy** [64, 65] is a *quality* (RQ2) that describes how confident individuals feel in each other's actual actions in a relationship (including communication, problem-solving, and supporting each other). Such confidence in the relationship naturally supports the building of effective metacommunicative practices. Compared to the AI-supported conditions (CAP, NCP), RWP participants, who experienced zero AI support, noticeably attributed more gains from the reflection process to their friends' actual behaviors, instead of to the system or to the overall conversation between the two friends.

This occurred in RWP participants despite the struggles most of them experienced from the lack of guidance compared to CAP or NCP: most of them explicitly reported difficulty even starting a reflective conversation about their interaction, and actively avoiding discussion of underlying feelings. All RWP participants mentioned that *"it was difficult"* (P2A) initiating the reflective discussion — *"keeping it natural was hard"* (P4A), *"it was hard to say what I thought"* (P21B), and *"I was thinking, like, what is that? What is understanding each other?"* (P11A). Many of them chose not to openly share their feelings during the reflection. P25B mentioned how they were unable to share their true feelings about their friend's desire to cheat on his partner:

"During the discussion part, I felt like I was good at underplaying it. Like, 'Oh, that shocked me. Haha!,' sounds more mild than, 'It actually shocked me a little bit that you're this kind of person.' I didn't say, 'Oh, this shocked me. I will think about this for the next two weeks and when I see you.'"

They employed avoidant strategies during the reflection such as saying *"I didn't feel anything uncomfortable"* despite revealing they experienced such moments in the interview (P11A). Instead of addressing or reflecting on points from their conversation, many RWP participants chose to continue their discussion of the conversation topic. For many, it felt like *"continuation of the conversation,"* (P20A), but *"deeper than what we usually talk about."* (P9B)

Although many RWP participants were not able to talk about the problems they experienced in the initial conversations, a number of them highlighted feelings of care, closeness, and strength in their friendship which they attributed to *specific behaviors from their friend* that made them feel this way. Several RWP participants mentioned feeling reassured and relieved that they were *"close enough"* with their friend to have such a conversation at all (P9B, P4AB, P11B, P20B, P25B, P27A). Though such feelings were also reported by NCP participants (Section 5.1.2), RWP participants explained a much higher amount of detail about their friend's behaviors, instead of attributing their insights to the discussion or the poem (which is what we observed in the other conditions). These participants described developing insights after their reflection specifically because of 'what their friend did,' versus 'what happened during the conversation.' For example, P9A described how her friend's behavior made her feel more empathy for her friend:

*"I think at first, I thought her as more of, not a very emotional person. **But after she talked** about how she felt uncomfortable with [redacted], that part really made me feel like oh, she's also pretty emotional and she compares herself to other people as well."*

Similarly, P20B described being comforted by his friend's behavior during the reflection:

*"I feel much better knowing that he doesn't feel annoyed and is always prepared to listen to my problems. I asked more detail about how he felt and **it was the way he answered**, that makes me feel like, 'Oh yeah. He really cares about me. He's very concerned about my well-being. Usually, he doesn't say those things.'"*

Others also used phrases such as “*hearing that he was like*,” (P30B) or “*the way he acted*” (P25B) to describe interactions with their friend during the reflection which clarified to them what kind of person their friend was, such as “*a more caring person*” (P30B) than they had thought. Participants reflected a sense of **relational efficacy** and confidence in their friend by explicitly attributing feelings of relationship closeness to their friend’s behaviors instead of to the conversation.

These RWP participants mentioned the importance of actively initiating more reflective conversations with each other in the future, which is an important metacommunicative practice. Although many RWP participants took away more generalized insights, such as knowledge of “*cultural aspects*” (P9B), or how to deal with a specific situation (P21A), others mentioned wanting to have more reflective conversations in the future. P21B mentioned getting her friend to “*open up*” through these reflections, while P25A thought “*reflective chats are important when I feel a little uncomfortable*.” P20B how they are necessary to help remind him about aspects important to their friendship:

“I can talk about anything with him. I already know that, but we never know when there might be a change of mind, unless we talk about this. So I think it’s better if, reflectively, we just tell each other constantly, to remind each other that, yes, it’s okay, we’ll always be here, and yes, you can talk to me.”

Though RWP participants were asked to reflect with no other support, the mere presence of an explicit space to reflect had a significant impact for them. RWP participant P27A mentioned,

“This platform makes you think at the moment what went wrong or right with the conversation. I think that if there isn’t a reflective discussion I wouldn’t go through the conversation and think about it unless it was really serious.”

P30B, who was usually reticent and whose friend was surprised at how much he expressed during the reflection, explained that he “*just needed a place to ask and express my thoughts, gave me the chance*.” The appreciation for a space to talk was not reflected only by RWP participants, but participants across all conditions. NCP P29B described the benefit of the reflective space: “*In this situation, we can talk more easily. Sometimes we worry about how something that is right in one culture may not be right in another, so we can’t talk about it. But in this system we can*.”

Overall, we observed that RWP participants (who were instructed to reflect without any support) did experience difficulties initiating reflection on their prior interactions and resolving specific metacommunicational issues they had experienced. However, compared to other conditions, we saw that many RWP participants particularly attributed post-reflection feelings of closeness to their friends’ behavior, instead of to the conversation or the system. We relate this to a sense of perceived **relational efficacy** due to the confidence these participants showed towards their friends’ behavior and their motivation to engage in more reflective conversations with each other.

5.3 Divergent Effects of AI-generated Content on the Social Reflection Process

Aside from the presence of a reflective space, the presence of AI also had significant and divergent impacts on the reflective conversation *process* (RQ1) between friends. Participants across all conditions reported complex, conflicting reactions to encountering AI elements of the system. This included contrasts in participants’ responses to the presence of AI and their trust in AI. Participant P11A (RWP) explained a feeling of invasiveness from the perceived presence of AI, although they only received a simple instruction to reflect in the chat:

“It’s because AI just came into our conversation and said it. If it was a real person doing the same thing that’s fine, but suddenly I feel like someone who doesn’t know our conversation just came in and organized our private conversation. I know the AI doesn’t think about anything. But it’s like we’re supervised - we are seen. It feels creepy.”

Even CAP and NCP participants, who experienced the support of AI-generated poems, had divergent reactions as well. P26B (CAP) also reported feeling as if *“there were a third person observing our chat,”* but instead described a sense of comfort in the presence of that entity as *“a neutral party”* which made them more comfortable to speak freely during the reflection. In contrast, P5B (CAP) expressed a sense of distrust in the AI’s ability to generate affective content, explaining: *“First of all, [the poem] is AI-generated, and it leads me to question whether we need to take it seriously. Artistically speaking, I didn’t feel it was written in a very beautiful way.”*

Such conflicting responses to the AI not only stemmed from doubt towards it, but also feelings of faith in the AI and lack of control. Some participants reported feeling *“bound to the AI”* due to an involuntary sense of faith in its generated content — P24A (NCP) described how *“when I see the poem and those [AI-generated] words, it started running in my mind.”* Due to this, they experienced difficulty thinking and discussing more freely during the reflection. While some participants distrusted the AI content and others felt restricted by their sense of trust in it, others still responded positively to trusting the AI content. P14B (CAP) remarked on how her sense of belief in what she saw as an AI-generated affirmation of their conversation made reflecting easier:

“Having this [poem] was a bit more liberating for me, to talk more. Thinking, that if the program can analyze our language and make a conclusion that is close to our conversation, then that means we also understand each other.”

These highly contrasting participant reactions to AI presence and their trust in the generated content demonstrate the complexity of AI effects in a personal interactive setting. Some participants felt more welcome to speak and validated about their friendship, while others felt constrained.

We also observed that varying quality in the AI-generated output for NCP and CAP conditions had different effects on participants’ reflections. Although we purposefully generated poems (and poem-based images) which are naturally open to interpretation, participants still encountered some cases of overly *“confusing”* or *“ambiguous”* output. For example, P13B reported how *“[the poem] works well when I see parts that resonate with things we said, but it’s often a mixed feeling because I don’t understand everything,”* and P28B recalled feeling like *“What am I looking at?”* in response to an image with ambiguous spatial orientation. P5B commented that one of their poems could have contained more *“emotion words”* instead of ambiguous lines like *‘questions of the path,’* while P7B stated that *“if the image was more concrete, like a mirror or a rock, rather than something vague, it would be easier to connect to the conversation.”*

Although overly ambiguous content was more often a hindrance, they were still helpful for reflective discussions in some cases. Most participants commented on the images’ content being confusing, but often emphasized their emotional appeal, such as images’ *“color palette”* (P22B), *“friendly”* (P8A), and *“gentle”* (P14B) qualities. These aesthetic features set a personal tone for the joint reflection. Some participants also drove their reflective discussion by collaboratively critiquing the AI-generated material. P14A mentioned how even when *“both of us agree that the poem is not that accurate to reflect our conversation, we’ll also talk about that. Like, how we disagree with the poem.”* P28A similarly expressed viewing the poem as *“something that we read and we either agree or disagree.”* P14B also mentioned how one of the AI-generated images paired with the poem misrepresented the sexual orientation that was originally discussed, but that this gender bias reflected by the AI actually provided conversation material for her and P14A to talk about. This critical and open-ended perspective enabled participants to initiate meaningful discussions based on ambiguous parts of the AI-generated content, such as P6A:

“We talked about how our differences in humor with other Japanese people created friction in conversations. The AI summarized that as ‘connection through humor binds.’ I thought

it was going against what we said, regarding connection with other Japanese people. But for us, it was a connection. So I thought it was a really interesting line.”

The inconsistent nature of AI-generated output was confusing and distracting for most participants’ reflection task, but some participants leveraged the ambiguity to develop more interpretations of their own conversation and have more critically reflective discussions with one another.

Overall, we saw that AI-generated content and AI presence in personal reflections had very diverse effects on the effectiveness of reflective discussion. The same feature could have very contrasting effects, such as AI presence making a participant more comfortable to open up towards their friend or less comfortable with the personal conversation. And while some participants felt tied down by their trust in AI output, others naturally viewed it as something to think critically about. These perceptions also had different effects on the reflection depending on whether participants held negative or positive attitudes toward their trust (or lack of trust) in AI. Our discussion will elaborate on the implications these findings pose for the role of AI-mediated communication in personal relationship contexts.

Table 2. Principal Results — connection to RQs from qualitative findings.

RQ1: Poet-Weaver’s impact on the <i>process</i> of establishing metacommunication, a.k.a mutual understanding of communication practices (includes addressing communication failure)	Process of metacommunication • Context-specific AI-generated poems (CAP) drive pinpointing of specific conversational issues, while context-unaware poems (NCP) divert focus towards maintaining relationship positivity (Section 5.1) • Presence of AI-generated poems mediates awareness of friends’ behavioral impact on the relationship and sense of comfort to engage in challenging personal conversation (Sections 5.2, 5.3)
RQ2: Poet-Weaver’s impact on the <i>quality</i> of metacommunication and relationship between friends	Quality of metacommunication • CAP participants reported the highest number of resolved conversation issues (Table 1) and delineated specific metacommunication strategies for future interaction within their friendship (Section 5.1.1) • NCP participants reported increased general willingness to share information with their friend (Section 5.1.2) • RWP participants reported increased motivation to engage in active reflection on their friendship interactions (Section 5.2) Quality of relationship • NCP participants reported stronger sense of positive affinity after reflective discussions (Section 5.1.2) • RWP participants reported stronger sense of relational efficacy, confidence in friends’ behaviors reflecting positive views of relationship (Section 5.2)

6 Discussion

In this work, we designed the AI-generated poems in Poet-Weaver to support joint reflection on communication failure in personal relationships, and conducted a mixed-methods field study to evaluate its effect on intercultural friends’ ability to improve mutual understanding and metacommunication by addressing such failures. Through our findings, we did see that the poems generated by Poet-Weaver were helpful in supporting challenging and transformative discussions between

intercultural friends when subjective context was integrated. For the friends that benefited, it served as a neutral, openly-interpretable presence that provided a launching pad for understanding each other's perspectives more fearlessly. Despite this, we also saw that participants' friendship qualities such as growth (5.1) and ownership (5.2) were impacted by the context-sensitivity (5.1) and presence of Poet-Weaver's AI-generated output (5.3).

We follow by highlighting key issues for integrating AI-generated, context-sensitive relationship support in systems moving forward, such as when to push users toward relational growth (6.1), how to support person-centered responsibility for relationships during use of AIMC (6.2), and broader ethical issues to consider for the use of openly-interpretable AIMC in personal relationships (6.3).

6.1 When should we try to promote transformative growth in relationships, compared to reinforcing positive bonds?

Our findings showed the need for sensitivity to the context of the relationship state when deciding whether and when AIMC systems like Poet-Weaver should promote reflective discourse to help people adjust their perspectives towards the relationship (**transformative**), or support more comfortable activities such as bolstering positive emotional bonds already present in the relationship (**reinforcing**). For example, the context-aware poem (CAP) allowed users to bridge references to their issues of communication failure and better understand their differences in perspective, experiencing more **transformative** changes to their friendship. Although no-context poem (NCP) users experienced this less often, their **reinforcement** of positive emotional bonds to sidestep the discomfort of addressing communication issues is also beneficial for relationships [73].

Based on these findings, we suggest considerations for varying the stylistic qualities and context-reflectivity of AI-generated content when supporting challenging interactions with close others. Especially in the intercultural friendship context where anxiety and risks of misunderstanding can be high [21, 23, 106], it is important to consider whether AI as an explicit third-party presence should challenge the relationship to grow, or if it should reinforce existing positive aspects of the relationship (e.g., using positive *foregrounding* [68]). For example, if the relationship is in a volatile state because a communication failure touched upon a significant difference in values, supporting more positive bonding instead of poking at the context of the problem could be more beneficial at first. Bridges to talk about the problem can be provided later in time when individuals are calmer.

By collecting broad and temporally-relevant metrics such as the frequency and affective valence of recent conversations, AIMC systems could consider 'relationship state' when deciding whether to provide context cues that support more challenging transformative discussions about metacommunication issues, or instead support more positive supplementary relationship interactions. For example, if users' recent interactions have been frequent and positive before a communication issue came up, encouraging reflection through specific context cues embedded in AI-generated interpretations of the conversation may be appropriate. Conversely, if users interacted infrequently before encountering a communication failure, or if they had a series of conflicts, AI-generated interpretations can support more positive buffering through friendly or relaxing aesthetics in the generated content and refrain from pinpointing communication failure context to avoid exacerbating an already-shaky situation. The developmental stage of the relationship, such as how close the individuals are, may also be relevant to the timing of transformative interventions. In order to determine the right thresholds for implementing such context-specific AIMC support for challenging personal conflicts, we will need to further investigate users' differences in relationship values, orientations, and dynamics to minimize the risk of intervention.

6.2 AI's social presence can affect friends' attributions of responsibility and agency over their own relationship

Our findings across different levels of AI-mediated support, from no support to context-aware support (RWP, NCP, and CAP), showed how the degree of AI involvement impacted participants' sense of relational efficacy and responsibility over their own relationship — essentially, their sense of agency and ability to influence their relationship outcome. Given that prior work highlighted the communication challenges in intercultural friendships, we expected that participants who had to reflect their communication failures without any support from AI as a third presence (RWP) would have the hardest time succeeding. However, we saw that they actually displayed a higher level of relational efficacy by directly crediting their friends' behaviors to relationship improvements (see [Section 5.2](#)). Relational efficacy is important for developing effective communication skills and critical for resolving conflicts [64, 65], but we did not observe this in either of the conditions with AI-mediated support (NCP and CAP). Given that the attribution of behavioral responsibility was higher in conversation reflections without AI-mediated support, it is possible that the degree of AI-mediated support can obstruct certain holistic and important features of friendship maintenance.

When supporting challenging interactions within close, personal relationships, future AIMC systems should prioritize providing space for involved parties to attribute behavioral accomplishments to one another as people who are working on their communication *because they care about each other*, to prevent the consequences of people miscrediting the AI system for their own emotional labor. We saw some potential trends in this direction from how the reflections of Poet-Weaver users centered mostly around the AI-generated poem content (5.1), and although this resulted in successfully resolving communication problems, much of the friends' attention was centered around positively viewing the AI's support (5.3) instead of recognizing each other's efforts and communication skills (5.2). If left unchecked, this could lead towards a slippery slope into unnecessary (over)dependence on AI for maintaining relationships and developing social skills [28]. Long-term consequences of attributing human emotional labor to AI systems could mean that individuals may eventually be unable to distinguish their authentic sense of shared reality in interpersonal relationships, if they fail to properly attribute human behaviors that actually impact human-human relationships. This will negatively affect close relationships, which are only able to alleviate human loneliness through feeling mutually understood [80]. It is important for future studies to more closely investigate how different degrees and forms of AI support (e.g., long-term use, interchanging between mixed AIMC roles [39]) in a close relationship context can affect relational efficacy, potentially integrating qualitative probes with existing measures of relationship self-efficacy [62].

In current practice, one way to implement space for attributing friends' behaviors to each other is to purposefully adjust the frequency with which AIMC interventions are triggered in personal relationship contexts. Explicitly alternating between conflict instances where the reflective space but no AI support will be provided (e.g., RWP), and occasional instances of communication struggle where AI support will be provided (e.g., CAP) can allow users some clearly-defined space to perceive and acknowledge their friends' interactions more clearly, and avoid users completely depending on the system. Since our findings showed that providing the mechanics of a reflective space for users did not seem to diminish relational efficacy (Section 5.2), modulating how often AI should be present as a third-party can create space to make the core social actors in the relationship more visible. However, effective ways to adjust levels of AI presence in a long-term use context may depend on factors such as individual preferences, relationship dynamics, and the severity of the communication failure. This approach of temporally and situationally modulating the frequency of AI presence can also be helpful for supporting other relationship contexts, such as collaborating teams who need to develop a deeper collective understanding that can sustain over time.

6.3 Ethical points of consideration for engaging “openly-interpretable” AIMC in personal and reflective social contexts

Although we designed Poet-Weaver to minimize active engagement from AI and instead encourage open-ended interpretation through poetic stylization, we still observed cases of bias in both the AI output and participants’ responses to AI. Especially in this “openly-interpretable” use context where the inherent ‘uncertainty’ of Poet-Weaver’s AI-generated output confused some users (Section 5.3) and likely increased cognitive load, addressing these biases can be important for preventing AI overtrust and overreliance which is known to occur during overwhelming task demand [9].

For example, we observed confirmation bias [54] in participants, where they trusted the AI’s interpretation of their friendship interaction or felt involuntarily compelled to believe it (Section 5.3). This also links to recent reports of generative AI and ChatGPT dependency, where some users find it difficult to have independent thought in the presence of such technology [82]. This issue of reliance was not observed with the AI-generated image, which supported affective grounding [46] in the participants. However, our participants reported that some of the AI-generated images depicted gender biases, such as a heterosexual couple being used in the visual when the friends were actually discussing same-sex relationships (Section 5.3). Although the cases we observed only seemed to have positive effects on participants as creative conversation fodder, they still demonstrate a risk of injecting biases into the interaction if present in a context where users are overwhelmed and confused by the openly-interpretable content.

These ethical issues relate to prior work on concerning influences of AIMC for personal relationships, which showed tensions between trust in social decisions made by the AI [41, 110] and concern about overreliance on AIMC threatening the “*authenticity*” of interactions in personal relationships [28]. Our findings extend prior work on AIMC’s influence of individuals’ interpersonal evaluations by showing that users could end up trusting AI-generated evaluations of their relationship, which has mixed implications. External validation could be useful for developing a sense of cohesion and camaraderie within relationships [50, 107], but can also sabotage relationships depending on the accuracy of the AI evaluation. Additionally, cultural incongruencies and poor cross-cultural alignment in large language models [78, 95] make the issues of AI bias and misrepresentation more sensitive when applied in intercultural contexts where people have nuanced perspectives.

While our study shows that openly-interpretable AIMC can effectively support personal relationships, clear constraints and measures should be taken to facilitate users’ agency and their awareness of biases that may result from the presence of AI and the content it generates. In both our findings and prior work, users expressed a concern with the inability to exert agency over their own perceptions of relationship-level interactions due to feelings of involuntary reliance and trust in AI. This concern is enhanced when we acknowledge that relationship-level attributions guided by AI-generated content can be more difficult to reorient than direct attributions to individual people — due to fundamental attribution error, people have a strong tendency to focus on individual traits over the more “metacognitive” relational context [79]. But for the same reason, AI-generated evaluations of the broader relational context can also help users by providing perspectives that are difficult for them to form on their own, also shown in Section 5.3 of our work. To leverage the advantages of AIMC in personal relationships, future AIMC systems can combat the consequences of AI-generated evaluations at the relationship level by implementing mechanisms during the interpersonal reflection process that more actively prompt or encourage users to exercise accountability for relationship attributions they make [96]. One way to do this is to integrate a Socratic questioning process [20], which has been applied to encourage critical thinking in other AI use contexts, into personal AIMC. For example, the system can semantically detect moments during the social reflection process where interlocutors’ expressions reflect a change in

judgment, and explicitly ask the user “why” they developed this judgment. Special attention can be paid to users’ judgments or changes in perception that appear more closely related to larger social biases that are known to be reflected in generative AI, such as race and gender [38, 111]. In future work, experimenting with different forms of ‘open-endedness’ in AI-generated content, such as presenting well-formed questions that reference the conversation and request interlocutors to explain (e.g., “What did you two discuss about Christianity and childhood?”) instead of expressive, openly-interpretable content (the Poet-Weaver method) could also help enhance and diversify uses of AI to facilitate critical reflection at a relationship level.

7 Limitations and Future Work

While our quantitative measures did not reflect distinct differences across conditions, our qualitative findings demonstrated the complex and mixed effects AI-generated content had on a challenging, ‘high-stakes’ personal interaction. This context has seldom been explored in prior work and our findings show that people are guided by AI-generated content during co-reflective personal communication whether they like it or not. Further work should consider how the nature of AI content impacts different relational contexts in order to uncover how AIMC affects human relationships holistically (compared to humans as interacting individuals or members of a team). For example, perceptions of joint agency at the relationship level should be explored in addition to the more commonly-studied individual agency. Although our work’s focus on personal relationships may prioritize qualities like joint agency as important for sustaining bonds, strong social bonds are also important for the health of community relationships and organizational commitment in the work context [91, 100, 104]. As noted by Thomas [97], almost all relationships are intercultural to some degree and the ability to sustain bonds that can surpass differences in human perspective is globally essential for social health. Hence, the effectiveness of supporting difficult interactions with joint-use AIMC (where the AIMC faces both users at once, instead of each person using AI support in private) should also be studied in more depth and over a longer time to understand how this interaction model can impact patterns of relationship-level perception or behavior.

The challenges of studying communication failure in a semi-natural controlled setting also led to some limitations in our current work. As can be seen in the total counts of communication failures in Table 1, roughly two-thirds of the conversations analyzed for the study contained communication failures. Despite our efforts to help participants curate conversation topics that might lead to mismatches in understanding, we could not guarantee that all participants would experience communication failures since they happen unexpectedly and cannot be forced. Because our research question targeted participants’ ability to address communication failures, this may have impacted the power of our study and affected our ability to observe significant differences in the quantitative measures since not all participants had the chance to encounter and address them. Future controlled studies that wish to investigate serendipitously-occurring events such as communication failure may benefit from engaging in dynamic sampling until “saturation” of the event in question is reached, instead of setting the number of participants before recruitment.

8 Conclusion

Complex challenges in interpersonal communication have been supported by AI as social actors in various practical contexts. Personal interactions can also suffer from challenges that call for concrete social assistance, such as intercultural friendships’ struggle with lack of mutual understanding and suppressed communication. However, AI-mediated communication (AIMC) has a tenuous presence in the context of personal relationships due to ethical concerns about interfering with relationship authenticity and agency. In this work, we propose, design, and evaluate a system

which uses stylized, open-to-interpretation AI-generated discussion prompts as a form of minimal-engagement AIMC to support intercultural friends while preserving their agency in personal interactions. Poet-Weaver is a system which generates conversational summaries in the form of poems to support joint reflection on prior conversations. The generated poems integrate context from moments of conversational failure as hints for users to initiate challenging discussions about their communication issues. Our findings from a mixed-methods, controlled field study revealed that although the AI-generated poems embedded with conversational failure context did support participants in addressing communication problems, the social presence of AI-generated content still impacted participants' relationship views. We follow with future considerations for the role of AI in mediating personal relationship dynamics, weight, and identity.

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References

- [1] Amuseghan Sunday Adejisola. 2009. Language and communication in conflict resolution. *Journal of law and conflict resolution* 1, 1 (2009), 001–009. <https://academicjournals.org/journal/JLCR/article-full-text-pdf/EE4B4AB575>
- [2] Canfer Akbulut, Laura Weidinger, Arianna Manzini, Iason Gabriel, and Verena Rieser. 2024. All Too Human? Mapping and Mitigating the Risk from Anthropomorphic AI. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* 7, 1 (Oct. 2024), 13–26. doi:10.1609/aies.v7i1.31613 Number: 1.
- [3] Toshiki Aoki, Rintaro Chujo, Katsufumi Matsui, Saemi Choi, and Ari Hautasaari. 2022. EmoBalloon - Conveying Emotional Arousal in Text Chats with Speech Balloons. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3491102.3501920
- [4] Arthur Aron, Elaine N. Aron, and Danny Smollan. 1992. Inclusion of Other in the Self Scale and the structure of interpersonal closeness. *Journal of Personality and Social Psychology* 63, 4 (1992), 596–612. doi:10.1037/0022-3514.63.4.596 Place: US Publisher: American Psychological Association.
- [5] Hayet Bahri and Robert Williams. 2017. Interconnectivity and metacommunication. *Training Language and Culture* 1 (Sept. 2017), 8–29. doi:10.29366/2017tlc.1.3.1
- [6] Henri Barki and Jon Hartwick. 2004. CONCEPTUALIZING THE CONSTRUCT OF INTERPERSONAL CONFLICT. *International Journal of Conflict Management* 15, 3 (Jan. 2004), 216–244. doi:10.1108/eb022913 Publisher: Emerald Group Publishing Limited.
- [7] Genevieve Bell, Mark Blythe, and Phoebe Sengers. 2005. Making by making strange: Defamiliarization and the design of domestic technologies. *ACM Trans. Comput.-Hum. Interact.* 12, 2 (June 2005), 149–173. doi:10.1145/1067860.1067862
- [8] Staphord Bengesi, Hoda El-Sayed, MD Kamruzzaman Sarker, Yao Houkpati, John Irungu, and Timothy Oladunni. 2024. Advancements in Generative AI: A Comprehensive Review of GANs, GPT, Autoencoders, Diffusion Model, and Transformers. *IEEE Access* 12 (2024), 69812–69837. doi:10.1109/ACCESS.2024.3397775 Conference Name: IEEE Access.
- [9] Eric Bogert, Aaron Schecter, and Richard T. Watson. 2021. Humans rely more on algorithms than social influence as a task becomes more difficult. *Scientific Reports* 11, 1 (April 2021), 8028. doi:10.1038/s41598-021-87480-9 Publisher: Nature Publishing Group.
- [10] Ali Borji and Mehrdad Mohammadian. 2023. Battle of the Wordsmiths: Comparing ChatGPT, GPT-4, Claude, and Bard. doi:10.2139/ssrn.4476855
- [11] Virginia Braun and Victoria Clarke. 2012. Thematic analysis. In *APA handbook of research methods in psychology, Vol 2: Research designs: Quantitative, qualitative, neuropsychological, and biological*. American Psychological Association, Washington, DC, US, 57–71. doi:10.1037/13620-004
- [12] Emily A. Butler, Boris Egloff, Frank H. Wilhelm, Nancy C. Smith, Elizabeth A. Erickson, and James J. Gross. 2003. The social consequences of expressive suppression. *Emotion* 3, 1 (2003), 48–67. doi:10.1037/1528-3542.3.1.48 Place: US Publisher: American Psychological Association.
- [13] Fiona L. Calvert, Frank P. Deane, and Brin. F. S. Grenyer. 2020. Supervisee perceptions of the use of metacommunication in the supervisory relationship. *Psychotherapy Research* 30, 2 (Feb. 2020), 228–238. doi:10.1080/10503307.2018.1524169 Publisher: Routledge _eprint: <https://doi.org/10.1080/10503307.2018.1524169>
- [14] Yuh-Fang Chang. 2009. How to say no: an analysis of cross-cultural difference and pragmatic transfer. *Language Sciences* 31, 4 (July 2009), 477–493. doi:10.1016/j.langsci.2008.01.002

- [15] Jie Chen. 2015. Basic Theory Research on Pragmatic Failure for Intercultural Communication. Atlantis Press, 576–580. doi:10.2991/icemitt-15.2015.121 ISSN: 2352-5398.
- [16] Tiffany Chen, Cassandra Lee, Jessica R Mindel, Neska Elhaoui, and Rosalind Picard. 2023. Closer Worlds: Using Generative AI to Facilitate Intimate Conversations. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. Association for Computing Machinery, New York, NY, USA, 1–15. doi:10.1145/3544549.3585651
- [17] Guillem Collell Talleda and Marie-Francine Moens. 2016. Is an image worth more than a thousand words? On the fine-grain semantic differences between visual and linguistic representations. In *Proceedings of the 26th International Conference on Computational Linguistics*. ACL, 2807–2817. <https://lirias.kuleuven.be/1656676> ISSN: 9784879747020.
- [18] Shauna Concannon and Marcus Tomalin. 2024. Measuring perceived empathy in dialogue systems. *AI & SOCIETY* 39, 5 (Oct. 2024), 2233–2247. doi:10.1007/s00146-023-01715-z
- [19] Mauricio Cortina and Giovanni Liotti. 2010. Attachment is about safety and protection, intersubjectivity is about sharing and social understanding: The relationships between attachment and intersubjectivity. *Psychoanalytic Psychology* 27, 4 (2010), 410–441. doi:10.1037/a0019510 Place: US Publisher: Educational Publishing Foundation.
- [20] Valdemar Danry, Pat Pataranutaporn, Yaoli Mao, and Pattie Maes. 2023. Don't Just Tell Me, Ask Me: AI Systems that Intelligently Frame Explanations as Questions Improve Human Logical Discernment Accuracy over Causal AI explanations. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3544548.3580672
- [21] Kristin Davies, Linda R. Tropp, Arthur Aron, Thomas F. Pettigrew, and Stephen C. Wright. 2011. Cross-Group Friendships and Intergroup Attitudes: A Meta-Analytic Review. *Personality and Social Psychology Review* 15, 4 (Nov. 2011), 332–351. doi:10.1177/1088868311411103 Publisher: SAGE Publications Inc.
- [22] Jason D'Cruz, William Kidder, and Kush R. Varshney. 2021. The Empathy Gap: Why AI Can Forecast Behavior But Cannot Assess Trustworthiness.. In *TFSOCTAI@ AAAI Fall Symposium*. <https://ceur-ws.org/Vol-3332/paper9.pdf>
- [23] Karen K. Dion. 2023. Intergroup friendship: A reflective spotlight. *Personal Relationships* 30, 4 (2023), 1185–1207. doi:10.1111/per.12514 _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/per.12514>.
- [24] Wen Duan, Naomi Yamashita, and Susan R. Fussell. 2019. Increasing Native Speakers' Awareness of the Need to Slow Down in Multilingual Conversations Using a Real-Time Speech Speedometer. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (Nov. 2019), 171:1–171:25. doi:10.1145/3359273
- [25] Thomas Erickson and Wendy A. Kellogg. 2000. Social translucence: an approach to designing systems that support social processes. *ACM Trans. Comput.-Hum. Interact.* 7, 1 (March 2000), 59–83. doi:10.1145/344949.345004
- [26] Alan Fogel and Angela Uchoa Branco. 2014. Metacommunication as a source of indeterminism in relationship development. In *Dynamics and indeterminism in developmental and social processes*. Psychology Press, 65–92. <https://www.taylorfrancis.com/chapters/edit/10.4324/9781315806211-5/metacommunication-source-indeterminism-relationship-development-alan-fogel-angela-uchoa-branco>
- [27] Yue Fu, Yixin Chen, Zelia Gomes Da Costa Lai, and Alexis Hiniker. 2024. Should ChatGPT Write Your Breakup Text? Exploring the Role of AI in Relationship Dissolution. doi:10.48550/arXiv.2401.09695 arXiv:2401.09695.
- [28] Yue Fu, Sami Foell, Xuhai Xu, and Alexis Hiniker. 2024. From Text to Self: Users' Perception of AIMC Tools on Interpersonal Communication and Self. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–17. doi:10.1145/3613904.3641955
- [29] Fiona Fui-Hoon Nah, Ruilin Zheng, Jingyuan Cai, Keng Siau, and Langtao Chen. 2023. Generative AI and Chat-GPT: Applications, challenges, and AI-human collaboration. *Journal of Information Technology Case and Application Research* 25, 3 (July 2023), 277–304. doi:10.1080/15228053.2023.2233814 Publisher: Routledge _eprint: <https://doi.org/10.1080/15228053.2023.2233814>.
- [30] Nicholas Furlo, Jacob Gleason, Karen Feun, and Douglas Zytke. 2021. Rethinking Dating Apps as Sexual Consent Apps: A New Use Case for AI-Mediated Communication. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '21 Companion)*. Association for Computing Machinery, New York, NY, USA, 53–56. doi:10.1145/3462204.3481770
- [31] Leonarda García-Jimenez. 2014. The Pragmatic Metamodel of Communication: A cultural approach to interaction. *Studies in Communication Sciences* 14, 1 (Jan. 2014), 86–93. doi:10.1016/j.scoms.2014.03.006
- [32] Antonio García-Gómez. 2020. Intercultural and interpersonal communication failures: analyzing hostile interactions among British and Spanish university students on WhatsApp. *Intercultural Pragmatics* 17, 1 (March 2020), 27–51. doi:10.1515/ip-2020-0002 Publisher: De Gruyter Mouton.
- [33] Arthur M Glenberg and William E Langston. 1992. Comprehension of illustrated text: Pictures help to build mental models. *Journal of Memory and Language* 31, 2 (April 1992), 129–151. doi:10.1016/0749-596X(92)90008-L
- [34] Ella Glikson and Omri Asscher. 2023. AI-mediated apology in a multilingual work context: Implications for perceived authenticity and willingness to forgive. *Computers in Human Behavior* 140 (March 2023), 107592. doi:10.1016/j.chb.2022.107592

- [35] Saul Greenberg. 2001. Context as a Dynamic Construct. *Human-Computer Interaction* 16, 2-4 (Dec. 2001), 257–268. doi:10.1207/S15327051HCI16234_09 Publisher: Taylor & Francis _eprint: https://doi.org/10.1207/S15327051HCI16234_09.
- [36] Christiana Gregoriou. 2023. The linguistic levels of foregrounding in stylistics. In *The Routledge Handbook of Stylistics* (2 ed.). Routledge. Num Pages: 14.
- [37] Carla F. Griggio, Midas Nouwens, Joanna McGrenere, and Wendy E. Mackay. 2019. Augmenting Couples' Communication with Lifelines: Shared Timelines of Mixed Contextual Information. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300853
- [38] Nicole Gross. 2023. What ChatGPT Tells Us about Gender: A Cautionary Tale about Performativity and Gender Biases in AI. *Social Sciences* 12, 8 (Aug. 2023), 435. doi:10.3390/socsci12080435 Number: 8 Publisher: Multidisciplinary Digital Publishing Institute.
- [39] Jeffrey T Hancock, Mor Naaman, and Karen Levy. 2020. AI-Mediated Communication: Definition, Research Agenda, and Ethical Considerations. *Journal of Computer-Mediated Communication* 25, 1 (March 2020), 89–100. doi:10.1093/jcmc/zmz022
- [40] Jess Hohenstein and Malte Jung. 2020. AI as a moral crumple zone: The effects of AI-mediated communication on attribution and trust. *Computers in Human Behavior* 106 (May 2020), 106190. doi:10.1016/j.chb.2019.106190
- [41] Jess Hohenstein, Rene F. Kizilcec, Dominic DiFranzo, Zhila Aghajari, Hannah Mieczkowski, Karen Levy, Mor Naaman, Jeffrey Hancock, and Malte F. Jung. 2023. Artificial intelligence in communication impacts language and social relationships. *Scientific Reports* 13, 1 (April 2023), 5487. doi:10.1038/s41598-023-30938-9 Publisher: Nature Publishing Group.
- [42] Thomas Holtgraves. 1997. Styles of language use: Individual and cultural variability in conversational indirectness. *Journal of Personality and Social Psychology* 73, 3 (1997), 624–637. doi:10.1037/0022-3514.73.3.624 Place: US Publisher: American Psychological Association.
- [43] S. Hoppenbrouwers and H. Weigand. 2000. Meta-communication in the language-action perspective. In *Proceedings of the 5th International Workshop on the Language-Action Perspective on Communication Modelling (LAP2000)*, M. Schoop and C. Quix (Eds.). RWTH Aachen, Aachen, 131–149.
- [44] Haonan Hou, Kevin Leach, and Yu Huang. 2024. ChatGPT Giving Relationship Advice – How Reliable Is It? *Proceedings of the International AAAI Conference on Web and Social Media* 18 (May 2024), 610–623. doi:10.1609/icwsm.v18i1.31338
- [45] Maurice Jakesch, Megan French, Xiao Ma, Jeffrey T. Hancock, and Mor Naaman. 2019. AI-Mediated Communication: How the Perception that Profile Text was Written by AI Affects Trustworthiness. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300469
- [46] Malte F. Jung. 2017. Affective Grounding in Human-Robot Interaction. In *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction (HRI '17)*. Association for Computing Machinery, New York, NY, USA, 263–273. doi:10.1145/2909824.3020224
- [47] Malte F. Jung, Nikolas Martelaro, and Pamela J. Hinds. 2015. Using Robots to Moderate Team Conflict: The Case of Repairing Violations. In *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction (HRI '15)*. Association for Computing Machinery, New York, NY, USA, 229–236. doi:10.1145/2696454.2696460
- [48] Harold H. Kelley. 1986. Personal Relationships: Their Nature and Significance. In *The Emerging Field of Personal Relationships*. Routledge. Num Pages: 17.
- [49] Lynne Kelly and Aimee E. Miller-Ott. 2018. Perceived Miscommunication in Friends' and Romantic Partners' Texted Conversations. *Southern Communication Journal* 83, 4 (Aug. 2018), 267–280. doi:10.1080/1041794X.2018.1488271 Publisher: Routledge _eprint: <https://doi.org/10.1080/1041794X.2018.1488271>.
- [50] Blair Kidwell, Virginie Lopez-Kidwell, Christopher Blocker, and Erick M Mas. 2020. Birds of a Feather Feel Together: Emotional Ability Similarity in Consumer Interactions. *Journal of Consumer Research* 47, 2 (Aug. 2020), 215–236. doi:10.1093/jcr/ucaa011
- [51] Daun Kim, Ji Yeon Oh, Jae-Yeop Jeong, and Jin-Woo Jeong. 2023. EmoFlow: Visualizing Emotional Changes in Video Chat - Preliminary Study. In *Proceedings of the 25th International Conference on Mobile Human-Computer Interaction (MobileHCI '23 Companion)*. Association for Computing Machinery, New York, NY, USA, 1–7. doi:10.1145/3565066.3608702
- [52] Taewook Kim, Jung Soo Lee, Zhenhui Peng, and Xiaojuan Ma. 2019. Love in Lyrics: An Exploration of Supporting Textual Manifestation of Affection in Social Messaging. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW (Nov. 2019), 79:1–79:27. doi:10.1145/3359181
- [53] Mie Kito, Masaki Yuki, and Robert Thomson. 2017. Relational mobility and close relationships: A socioecological approach to explain cross-cultural differences. *Personal Relationships* 24, 1 (2017), 114–130. doi:10.1111/pere.12174 _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/pere.12174>.

- [54] Joshua Klayman. 1995. Varieties of Confirmation Bias. In *Psychology of Learning and Motivation*, Jerome Busemeyer, Reid Hastie, and Douglas L. Medin (Eds.). Vol. 32. Academic Press, 385–418. doi:10.1016/S0079-7421(08)60315-1
- [55] Hideaki Kuzuoka. 1992. Spatial workspace collaboration: a SharedView video support system for remote collaboration capability. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '92)*. Association for Computing Machinery, New York, NY, USA, 533–540. doi:10.1145/142750.142980
- [56] Hideaki Kuzuoka and Saul Greenberg. 1999. Mediating awareness and communication through digital but physical surrogates. In *CHI '99 Extended Abstracts on Human Factors in Computing Systems (CHI EA '99)*. Association for Computing Machinery, New York, NY, USA, 11–12. doi:10.1145/632716.632725
- [57] Jan Kučera, James Scott, Siân Lindley, and Patrick Olivier. 2021. Bedtime Window: A Field Study Connecting Bedrooms of Long-Distance Couples Using a Slow Photo-Stream and Shared Real-Time Inking. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3411764.3445121
- [58] Kuan-Yu Lin, Seraphina Yong, Shuo-Ping Wang, Chien-Tung Lai, and Hao-Chuan Wang. 2016. HandVis: Visualized Gesture Support for Remote Cross-Lingual Communication. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 1236–1242. doi:10.1145/2851581.2892431
- [59] Lydia Listyani. 2019. The Use of a Visual Image to Promote Narrative Writing Ability and Creativity. *Eurasian Journal of Educational Research* (2019). <https://eric.ed.gov/?id=EJ1211623> Publisher: ANI Publishing Company ERIC Number: EJ1211623.
- [60] Fannie Liu, Chunjong Park, Yu Jiang Tham, Tsung-Yu Tsai, Laura Dabbish, Geoff Kaufman, and Andrés Monroy-Hernández. 2021. Significant Otter: Understanding the Role of Biosignals in Communication. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–15. doi:10.1145/3411764.3445200
- [61] Xingyu "Bruce" Liu, Vladimir Kirilyuk, Xiuxiu Yuan, Alex Olwal, Peggy Chi, Xiang "Anthony" Chen, and Ruofei Du. 2023. Visual Captions: Augmenting Verbal Communication with On-the-fly Visuals. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–20. doi:10.1145/3544548.3581566
- [62] Frederick G. Lopez, Morúa , Wendy, , and Kenneth G. Rice. 2007. Factor Structure, Stability, and Predictive Validity of College Students' Relationship Self-Efficacy Beliefs. *Measurement and Evaluation in Counseling and Development* 40, 2 (July 2007), 80–96. doi:10.1080/07481756.2007.11909807 Publisher: Routledge _eprint: <https://doi.org/10.1080/07481756.2007.11909807>.
- [63] Howard J. Markman, Mari Jo Renick, Frank J. Floyd, Scott M. Stanley, and Mari Clements. 1993. Preventing marital distress through communication and conflict management training: A 4- and 5-year follow-up. *Journal of Consulting and Clinical Psychology* 61, 1 (1993), 70–77. doi:10.1037/0022-006X.61.1.70 Place: US Publisher: American Psychological Association.
- [64] Allyson L. McCurry, Paul Schrodtt, and Andrew M. Ledbetter. 2012. Relational uncertainty and communication efficacy as predictors of religious conversations in romantic relationships. *Journal of Social and Personal Relationships* 29, 8 (Dec. 2012), 1085–1108. doi:10.1177/0265407512449402 Publisher: SAGE Publications Ltd.
- [65] Suzanne Meeks, Diane B. Arnkoff, Carol R. Glass, and Clifford I. Notarius. 1986. Wives' Employment Status, Hassles, Communication, and Relational Efficacy: Intra- versus Extra-Relationship Factors and Marital Adjustment. *Family Relations* 35, 2 (1986), 249–255. doi:10.2307/583632 Publisher: [National Council on Family Relations, Wiley].
- [66] Morton J. Mendelson and Frances E. Aboud. 1999. Measuring friendship quality in late adolescents and young adults: McGill Friendship Questionnaires. *Canadian Journal of Behavioural Science / Revue canadienne des sciences du comportement* 31, 2 (1999), 130–132. doi:10.1037/h0087080 Place: Canada Publisher: Canadian Psychological Association.
- [67] David S. Miall and Don Kuiken. 1994. Foregrounding, defamiliarization, and affect: Response to literary stories. *Poetics* 22, 5 (Aug. 1994), 389–407. doi:10.1016/0304-422X(94)00011-5
- [68] Jan Mukarovský. 2000. Standard language and poetic language (1932). *Chapter 23* (2000), 225–230. https://books.google.com/books?hl=en&lr=&id=7S29-ASD1HgC&oi=fnd&pg=PA225&dq=foregrounding+mukarovsky&ots=fFmjDgk0w4&sig=RhTNx0z_Pf0NGF-8j-EkQktzsoc Publisher: The Routledge Language and Cultural Theory Reader.
- [69] Muhamad Bisri Mustofa and Siti Wuryan. 2024. Enhancing Interpersonal Communication through the Use of ChatGPT and AI: A Case Study on Improving Communication Quality in Online Learning Environments in Lampung. *AICCON* 1 (Feb. 2024), 118–124. <https://journal.rc-communication.com/index.php/AICCON/article/view/135>
- [70] Alexander O'Connor. [n. d.]. *Managing Intergroup Emotions: How Intergroup Ideologies and Emotion Regulation Can Stifle Positive Emotions and Intergroup Friendships*. Ph.D. University of California, Berkeley, United States – California. <https://www.proquest.com/docview/1114898285/abstract/C5E1CA056289438FPQ/1> ISBN: 9781267605269.

- [71] John Oetzel, Adolfo J. Garcia, and Stella Ting-Toomey. 2008. An analysis of the relationships among face concerns and facework behaviors in perceived conflict situations. *International Journal of Conflict Management* 19, 4 (Jan. 2008), 382–403. doi:10.1108/10444060810909310 Publisher: Emerald Group Publishing Limited.
- [72] Silviu Vlad Oprea and Walid Magdy. 2020. The Effect of Sociocultural Variables on Sarcasm Communication Online. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW1 (May 2020), 29:1–29:22. doi:10.1145/3392834
- [73] Nickola C. Overall, Jeffrey A. Simpson, and Helena Struthers. 2013. Buffering attachment-related avoidance: softening emotional and behavioral defenses during conflict discussions. *Journal of Personality and Social Psychology* 104, 5 (May 2013), 854–871. doi:10.1037/a0031798
- [74] Elizabeth Page-Gould, Rodolfo Mendoza-Denton, and Linda R. Tropp. 2008. With a little help from my cross-group friend: reducing anxiety in intergroup contexts through cross-group friendship. *Journal of Personality and Social Psychology* 95, 5 (Nov. 2008), 1080–1094. doi:10.1037/0022-3514.95.5.1080
- [75] Yi-Hao Peng, Ming-Wei Hsi, Paul Tael, Ting-Yu Lin, Po-En Lai, Leon Hsu, Tzu-chuan Chen, Te-Yen Wu, Yu-An Chen, Hsien-Hui Tang, and Mike Y. Chen. 2018. SpeechBubbles: Enhancing Captioning Experiences for Deaf and Hard-of-Hearing People in Group Conversations. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/3173574.3173867
- [76] Bernd Ploderer, Wolfgang Reitberger, Harri Oinas-Kukkonen, and Julia van Gemert-Pijnen. 2014. Social interaction and reflection for behaviour change. *Personal and Ubiquitous Computing* 18, 7 (Oct. 2014), 1667–1676. doi:10.1007/s00779-014-0779-y
- [77] Michael Polanyi. 2015. *Personal Knowledge: Towards a Post-Critical Philosophy*. University of Chicago Press, Chicago, IL. <https://press.uchicago.edu/ucp/books/book/chicago/P/bo19722848.html>
- [78] Vinodkumar Prabhakaran, Rida Qadri, and Ben Hutchinson. 2022. Cultural Incongruencies in Artificial Intelligence. doi:10.48550/arXiv.2211.13069 arXiv:2211.13069 [cs].
- [79] Lee Ross. 1977. The Intuitive Psychologist And His Shortcomings: Distortions in the Attribution Process1. In *Advances in Experimental Social Psychology*, Leonard Berkowitz (Ed.). Vol. 10. Academic Press, 173–220. doi:10.1016/S0065-2601(08)60357-3
- [80] Maya Rossignac-Milon and E Tory Higgins. 2018. Epistemic companions: shared reality development in close relationships. *Current Opinion in Psychology* 23 (Oct. 2018), 66–71. doi:10.1016/j.copsyc.2018.01.001
- [81] Kristin Rygg. 2012. *Direct and indirect communicative styles : a study in sociopragmatics and intercultural communication based on interview discourse with Norwegian and Japanese business executives*. Doctoral thesis. Universitetet i Bergen. <https://openaccess.nhh.no/nhh-xmlui/handle/11250/164267> Accepted: 2012-06-18T10:08:52Z ISBN: 9788230820315.
- [82] Mohammed Salah, Fadi Abdelfattah, Hussam Alhalbusi, and Muna Al Mukhaini. 2024. Me and My AI Bot: Exploring the 'Alholic' Phenomenon and University Students' Dependency on Generative AI Chatbots - Is This the New Academic Addiction? doi:10.21203/rs.3.rs-3508563/v2 ISSN: 2693-5015.
- [83] Morgan Sandler, Hyesun Choung, Arun Ross, and Prabu David. 2024. A Linguistic Comparison between Human and ChatGPT-Generated Conversations. doi:10.48550/arXiv.2401.16587 arXiv:2401.16587.
- [84] Hiroaki Sato. 1987. Lineation of Tanka in English Translation. *Monumenta Nipponica* 42, 3 (1987), 347–356. doi:10.2307/2384931 Publisher: Sophia University.
- [85] Anna Schouten, Michael Boiger, Alexander Kirchner-Häusler, Yukiko Uchida, and Batja Mesquita. 2020. Cultural Differences in Emotion Suppression in Belgian and Japanese Couples: A Social Functional Model. *Frontiers in Psychology* 11 (2020). <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2020.01048>
- [86] Sarah Sebo, Brett Stoll, Brian Scassellati, and Malte F. Jung. 2020. Robots in Groups and Teams: A Literature Review. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW2 (Oct. 2020), 176:1–176:36. doi:10.1145/3415247
- [87] Elena Semino. 1995. Schema theory and the analysis of text worlds in poetry. *Language and Literature* 4, 2 (May 1995), 79–108. doi:10.1177/096394709500400201 Publisher: SAGE Publications Ltd.
- [88] Ashish Sharma, Inna W. Lin, Adam S. Miner, David C. Atkins, and Tim Althoff. 2023. Human-AI collaboration enables more empathic conversations in text-based peer-to-peer mental health support. *Nature Machine Intelligence* 5, 1 (Jan. 2023), 46–57. doi:10.1038/s42256-022-00593-2 Publisher: Nature Publishing Group.
- [89] Solace Shen, Petr Slovak, and Malte F. Jung. 2018. "Stop. I See a Conflict Happening.": A Robot Mediator for Young Children's Interpersonal Conflict Resolution. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction (HRI '18)*. Association for Computing Machinery, New York, NY, USA, 69–77. doi:10.1145/3171221.3171248
- [90] Lei Shi, Brianna J. Tomlinson, John Tang, Edward Cutrell, Daniel McDuff, Gina Venolia, Paul Johns, and Kael Rowan. 2019. Accessible Video Calling: Enabling Nonvisual Perception of Visual Conversation Cues. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW (Nov. 2019), 131:1–131:22. doi:10.1145/3359233
- [91] Randi L. Sims. 2002. Ethical Rule Breaking by Employees: A Test of Social Bonding Theory. *Journal of Business Ethics* 40, 2 (Oct. 2002), 101–109. doi:10.1023/A:1020330801847

- [92] Helen Spencer-Oatey. 2002. Managing rapport in talk: Using rapport sensitive incidents to explore the motivational concerns underlying the management of relations. *Journal of Pragmatics* 34, 5 (May 2002), 529–545. doi:10.1016/S0378-2166(01)00039-X
- [93] Minhyang (Mia) Suh, Emily Youngblom, Michael Terry, and Carrie J Cai. 2021. AI as Social Glue: Uncovering the Roles of Deep Generative AI during Social Music Composition. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3411764.3445219
- [94] Michael Sun and Anna S. Lau. 2018. Exploring Cultural Differences in Expressive Suppression and Emotion Recognition. *Journal of Cross-Cultural Psychology* 49, 4 (May 2018), 664–672. doi:10.1177/0022022118763749 Publisher: SAGE Publications Inc.
- [95] Yan Tao, Olga Viberg, Ryan S Baker, and René F Kizilcec. 2024. Cultural bias and cultural alignment of large language models. *PNAS Nexus* 3, 9 (Sept. 2024), pgae346. doi:10.1093/pnasnexus/pgae346
- [96] Philip E. Tetlock. 1985. Accountability: A Social Check on the Fundamental Attribution Error. *Social Psychology Quarterly* 48, 3 (1985), 227–236. doi:10.2307/3033683 Publisher: [Sage Publications, Inc., American Sociological Association].
- [97] Jenny Thomas. 1983. Cross-Cultural Pragmatic Failure. *Applied Linguistics* 4, 2 (July 1983), 91–112. doi:10.1093/applin/4.2.91
- [98] Bill Thompson, Seán G. Roberts, and Gary Lupyan. 2020. Cultural influences on word meanings revealed through large-scale semantic alignment. *Nature Human Behaviour* 4, 10 (Oct. 2020), 1029–1038. doi:10.1038/s41562-020-0924-8 Number: 10 Publisher: Nature Publishing Group.
- [99] Metin Timuçin. 2010. EXPLORING THE LANGUAGE OF POEMS: A STYLISTIC STUDY. *Novitas-Royal* 4, 2 (2010). <https://www.academia.edu/download/33321216/stylisticAnlsyz.pdf>
- [100] Bahar Tunçgenç, Valerie van Mulukom, and Martha Newson. 2023. Social bonds are related to health behaviors and positive well-being globally. *Science Advances* 9, 2 (Jan. 2023), eadd3715. doi:10.1126/sciadv.add3715 Publisher: American Association for the Advancement of Science.
- [101] Andrea Tyler. 1995. The Coconstruction of Cross-Cultural Miscommunication: Conflicts in Perception, Negotiation, and Enactment of Participant Role and Status Andrea Tyler. *Studies in Second Language Acquisition* 17, 2 (June 1995), 129–152. doi:10.1017/S0272263100014133
- [102] Jacquie D. Vorauer and Yumiko Sakamoto. 2006. I Thought We Could Be Friends, but . . . : Systematic Miscommunication and Defensive Distancing as Obstacles to Cross-Group Friendship Formation. *Psychological Science* 17, 4 (April 2006), 326–331. doi:10.1111/j.1467-9280.2006.01706.x Publisher: SAGE Publications Inc.
- [103] Bryan Wang, Yuliang Li, Zhaoyang Lv, Haijun Xia, Yan Xu, and Raj Sodhi. 2024. LAVE: LLM-Powered Agent Assistance and Language Augmentation for Video Editing. In *Proceedings of the 29th International Conference on Intelligent User Interfaces (IUI '24)*. Association for Computing Machinery, New York, NY, USA, 699–714. doi:10.1145/3640543.3645143
- [104] Edward Shih-Tse Wang. 2014. Moderators of the relationship between social bonding and organizational commitment. *Managing Service Quality* 24, 3 (May 2014), 300–314. doi:10.1108/MSQ-08-2013-0157 Publisher: Emerald Group Publishing Limited.
- [105] Meifen Wei, Jenny C. Su, Stephanie Carrera, Shu-Ping Lin, and Fei Yi. 2013. Suppression and interpersonal harmony: A cross-cultural comparison between Chinese and European Americans. *Journal of Counseling Psychology* 60, 4 (2013), 625–633. doi:10.1037/a0033413 Place: US Publisher: American Psychological Association.
- [106] Tessa V. West. 2011. Interpersonal perception in cross-group interactions: Challenges and potential solutions. *European Review of Social Psychology* 22, 1 (March 2011), 364–401. doi:10.1080/10463283.2011.641328 Publisher: Routledge _eprint: <https://doi.org/10.1080/10463283.2011.641328>.
- [107] Tessa V. West, Joe C. Magee, Sarah H. Gordon, and Lindy Gullett. 2014. A little similarity goes a long way: the effects of peripheral but self-revealing similarities on improving and sustaining interracial relationships. *Journal of Personality and Social Psychology* 107, 1 (July 2014), 81–100. doi:10.1037/a0036556
- [108] Hisako Yamagashira. 2001. Pragmatic transfer in Japanese ESL refusals. *Kagoshima Immaculate Heart College, English Department* (2001), 259–275. https://k-junshin.repo.nii.ac.jp/record/545/files/kiyo_31_15.pdf
- [109] Muneo Jay Yoshikawa. 1987. The double-swing model of intercultural communication between the East and the West. In *Communication theory*. Elsevier, 319–329.
- [110] Guanglu Zhang, Leah Chong, Kenneth Kotovsky, and Jonathan Cagan. 2023. Trust in an AI versus a Human teammate: The effects of teammate identity and performance on Human-AI cooperation. *Computers in Human Behavior* 139 (Feb. 2023), 107536. doi:10.1016/j.chb.2022.107536
- [111] Mi Zhou, Vibhanshu Abhishek, Timothy Derdenger, Jaymo Kim, and Kannan Srinivasan. 2024. Bias in Generative AI. doi:10.48550/arXiv.2403.02726 arXiv:2403.02726 [econ].

A Pilot Study

A.1 Rationale, Design, and Procedure

We recruited five pairs of friends (only one user overlapped with the pool recruited for the interview probe) to conduct preliminary evaluation on the two styles of AI-generated conversation summaries (see [Section 3.1.2](#)).

Rationale and Design. Our evaluation targeted the consistency and level to which each type of AI-generated summary was able to support the behavioral goals of our system: improve the ability of a user who has experienced communication failure in voicing and addressing the issue with their friend. We targeted our evaluation of the quality of these summaries based on whether and how well they were able to support this process, which included: specifically noting whether users were able to identify the indirect references to subjective context injected into the AI-generated summaries, the users' perceptions of the summaries in relation to their conversation and friendship, and how well the users were able to describe concretely their use of the summary to address the real conversation failure from the text conversation they had with their friend.

Because our AI-generated summaries described the social experiences of the interlocutors during their conversation (instead of directly summarizing the textual content of the conversation), these summaries are inherently subjective as there is no ground truth for a 'correct' summary of a social experience; there can be significant and debatable variation over which intangible social elements should be included and to what degree. Recent work has also articulated the difficulty and lack of standard process in evaluating performance metrics such as trustworthiness for AI representations of human perspective [18, 22]. For this reason, we focus on evaluating the contextual applicability of the AI-generated summary in lieu of measuring objective performance qualities. However, we do measure the identifiability of privately-marked conversation segments in the AI-generated summaries as a partial quantitative metric of summary quality (see [A.2](#)).

Procedure and Analysis. We collected the transcripts of two 15-minute text conversations from each pair of friends where each person had privately marked messages with perceived communication failures. We generated both types of conversation summaries (plain-language and poem) based on these real conversations and conducted interviews with each participant to specifically probe how they would react to using each summary to reflect on their conversation and address the communication failures they experienced. This interview process was repeated with each user individually, for all four summaries per person (two types of summaries per conversation).

Our interview questions were structured to evaluate specifically whether each participant was able to identify the conversational context of communication failures they had marked inside the generated summaries, as well as how the linguistic style of the AI-generated conversation prompt affected perceptions of directness, tone, and agency in the interaction. During analysis, the first author denoted which summaries did/did not have user-recognizable references to privately marked conversation segments, open-coded the interviews with each participant, and conducted thematic analysis to generate our preliminary findings.

A.2 Preliminary Findings

Across ten users (five friendships), participants were able to clearly identify their privately-marked segments of communication failure in 35/40 poem-format summaries and 38/40 plain-language summaries. However, overall we found that nearly all participants preferred the poem over the plain-language conversation summary, and that the image visually-aided interpreting the poem. Therefore, we decided to implement the stylized conversational summary as a poem and image in our system ([Section 3.2](#)) due to the small difference across summaries in identifiability of subjective context and the comparative conversational advantage of the poem summary, which will be

described below. Main themes discussed by participants included ease of comprehension, affective approach towards the conversation, and perceived cognitive agency.

Comprehension. Participants remarked that the poem was, overall, easier to understand and better supported them to think of ways that they could open up a conversation about the communication failure with their friend. Most participants mentioned that the length of the poem made it easier to understand, and as P2 mentioned, *“Because the plain-language summary is more specific and not 100% correct, the parts that feel misrepresented stand out more.”* In contrast, P3 explained how *“the poem is a bit abstract at first but if I spend a bit of time to interpret it with my friend, it’s a better conversation starter.”* It was also easier for all participants to identify phrases or lines of the poem that corresponded to the parts of their prior conversations where they marked communication failures.

Affect. The poem was also favored by participants for appealing to the personal nature of their conversations with their friends. P10 remarked that it felt as if the poem goes a bit *“deeper”* and used the analogy, *“The [plain-language] summary is like a recipe, and the poem is like the smell of the cake you baked.”* P9 mentioned how the plain-language summary’s use of conceptual terms like ‘societal’ *“feels abstract and disconnected from the personal experiences we were sharing,”* and P5 mentioned that it felt *“mechanical.”*

Agency. Although the plain-language conversation summary was more representative of a straightforward response from ChatGPT, participants preferred it less due to feeling as if their agency was infringed upon. P5 remarked that they felt plain-language summary was *“directing or instructing me on what to do.”* P9 made a similar statement on how the specific details provided about their conversation in plain-language *“made it more difficult to redirect the conversation”* to address what she needed to talk about, while *“the poem has no direction, it’s just words and concepts so it’s easy to direct it where we want to go.”*

Visual Aid. Participants who struggled to interpret the abstract language in the poem found the image helpful. For example, P7 struggled at first to see the connection between the poem and the conversation topic, but *“when I see the picture it’s easier for me to associate with more ideas that are potentially related to the poem.”* They cited a specific case where they did not think about the social context of the poem until they saw people interacting with each other in the generated image.

B GPT-4o and DALLÉ-3 Prompts

Notes: During implementation and testing, we had to repeat certain instructions several times for GPT-4o to not mention parts of the conversation directly in order to fulfill the indirect requirement of our system design, partly contributing to the length of the prompt.

B.1 Prompt for Poet-Weaver Context-aware Poem (CAP)

You are an assistant who will generate concise summaries of formatted text conversations presented to you while following each of the instructions described here. Marked messages in the conversation appended with (marked) indicate that the recipient of that message feels either confusion, mixed or negative feelings upon reading it. In context of the overall conversation, please interpret the social themes and personal experiences implied by: (1) why the recipient may have marked the message, and (2) how the recipient responded to the message, in context of difficulties encountered in cross-cultural personal relationships (please don't output your interpretation of these themes, only the summary). After you interpret these themes, generate a summary of the conversation that fulfills the following criteria. Your summary should describe in a general/conceptual sense what happened in the conversation through a lens that pays particular attention to social themes of the interaction and each person's experience during it. Your summary should highlight themes you interpreted from the marked messages. Your summary needs to be so abstract that specific parts of the conversation cannot be identified by either speaker through reading the summary. When you want to mention a part of the conversation relating to a marked message, please use keywords different from those used in the messages themselves. Your summary cannot mention specific parts of the conversation in any way, including anything about message marking, either of the individual speakers, or directly mentioning their individual feelings or experiences during the conversation. Your summary should be addressed to the people who had the conversation you summarized and maintain a tone that is more social and not too mechanical. Do not mention cultural differences. Please allow your interpretation to stay close to the text. Please turn your summary into a tanka poem that highlights the implications of the marked messages.

B.2 Prompt for Poet-Weaver No-context Poem (NCP)

You are an assistant who will generate concise summaries of formatted text conversations presented to you while following each of the instructions described here. In context of the overall conversation, please interpret the social themes and personal experiences implied by the conversation these two people had. After you interpret these themes, generate a summary of the conversation that fulfills the following criteria. Your summary should describe in a general/conceptual sense what happened in the conversation through a lens that pays particular attention to social themes of the interaction and each person's experience during it. Your summary needs to be so abstract that specific parts of the conversation cannot be identified by either speaker through reading the summary. When you want to mention a part of the conversation, please use keywords different from those used in the messages themselves. Your summary cannot mention specific parts of the conversation in any way, including either of the individual speakers, or directly mentioning their individual feelings or experiences during the conversation. Your summary should be addressed to the people who had the conversation you summarized and maintain a tone that is more social and not too mechanical. Do not mention cultural differences. Please allow your interpretation to stay close to the text. Please turn your summary into a tanka poem that highlights the implications of the conversation.

B.3 DALLE-3 Image Generation Prompt

Focus on specific, visually representable elements. Describe actions and scenarios rather than abstract concepts. Avoid ambiguous language that could be interpreted as including text. Create an imagery-only (no text!) visual of a person's experience in the following poem, in the style of minimalist and surreal illustration: poem entered here

C Participant Demographics

Table 3. Demographic data of participating friends in system evaluation study.

Pair ID	Age/Gender (Friend A)	Age/Gender (Friend B)	Differences in cultural and personal background which affect their interaction
1	20 F	20 F	Raised in rural vs. urban area, differences in socioeconomic status
2	21 F	21 F	Differences in family and academic background
3	24 M	27 M	Cross-national cultural differences; born and raised in different countries
4	33F	32 M	Cross-national cultural differences; born and raised in different countries
5	22 M	25 F	Cross-national cultural differences; born and raised in different countries
6	22 F	20 M	Cross-national cultural differences; born and raised in different countries
7	33 M	26 M	Age gap, cross-national cultural differences; born and raised in different countries
8	23 NB	27 F	Cross-national cultural differences; born and raised in different countries
9	21 F	22 F	Cross-national cultural differences; born and raised in different countries
10	26 M	28 M	Cross-national cultural differences; born and raised in different countries
11	23 F	24 F	Cross-national cultural differences; born and raised in different countries
12	28 F	28 M	Religion, socioeconomic status, cross-national cultural differences
13	21 M	23 F	Cross-national cultural differences; born and raised in different countries
14	27 F	32 F	Cross-national cultural differences; born and raised in different countries
15	28 F	25 F	Age gap, religion, cross-national cultural differences
16	23 F	22 F	Differences in family upbringing, academic background, and personality
17	31 F	23 F	Cross-national cultural differences; born and raised in different countries
18	30 M	26 M	Differences in socioeconomic status, family upbringing, and academic background
19	31 F	26 F	Cross-national cultural differences; born and raised in different countries
20	31 M	30 M	Differences in ethnicity and religion
21	32 F	26 F	Differences in family, community of upbringing and social beliefs
22	22 F	24 F	Cross-national cultural differences; born and raised in different countries
23	26 M	28 M	Cross-national cultural differences; born and raised in different countries
24	23 F	23 M	Cross-national cultural differences; born and raised in different countries
25	24 M	23 NB	Cross-national cultural differences; born and raised in different countries
26	32 F	28 M	Differences in family relationships, cross-national cultural differences; born and raised in different countries
27	24 F	30 M	Differences in ethnicity, religion, and family background
28	32 M	31 M	Differences in ethnicity and religion
29	24 F	26 F	Differences in community of upbringing, cross-national cultural differences; born and raised in different countries
30	38 M	37 M	Differences in familial and political environment of upbringing; socially liberal and diverse vs. socially conservative

D Questionnaire Items

The McGill Friendship Questionnaire - Respondent's Affection (MFQ-RA) [66] has two subscales: *Satisfaction* (S) and *Positive Feelings* (P). Respondents rate agreement with each item on a 9-point scale from "Very Much Disagree" to "Very Much Agree." Our short form retained the same ratio of subscale items in the original but reduced the number of questions from 16 to 9.

- (1) I am happy with my friendship with _____. (S)
- (2) I care about _____. (P)
- (3) I feel my friendship with _____ is a great one. (S)
- (4) I want to stay friends with _____ for a long time. (P)
- (5) I feel close to _____. (P)
- (6) I think my friendship with _____ is strong. (S)
- (7) I am glad that _____ is my friend. (P)
- (8) I would miss _____ if they left (P)
- (9) I am contented with my friendship with _____. (S)

The McGill Friendship Questionnaire - Friend's Function (MFQ-FF) [66] assesses six friendship functions (stimulating companionship, help, intimacy, reliable alliance, self-validation, and emotional security) in 30 items. We developed a 10-item short form to assess the functions most relevant to the types of interactions participants engaged in for our study: *Intimacy* (I), *Reliable Alliance* (R), and *Emotional Security* (E). Each item lists a statement about the friend in question, and respondents rate their level of agreement on a 9-point scale from "Never" to "Always."

- (1) _____ would make me feel comfortable in a new situation. (E)
- (2) _____ is someone who I can tell private things to. (I)
- (3) _____ would want to stay my friend if we didn't see each other for a few months. (R)
- (4) _____ would still want to be my friend even if I were angry with them. (R)
- (5) _____ would make me feel better if I were worried. (E)
- (6) _____ would still want to be my friend even if we had a fight. (R)
- (7) _____ knows when something bothers me. (I)
- (8) _____ would stay my friend even if other people did not like me. (R)
- (9) _____ would make me feel calmer if I were nervous. (E)
- (10) _____ is easy to talk to about private things. (I)

The Metacommunication in Supervision Questionnaire (MSQ) [13] is a 12-item scale that rates the quality and willingness of metacommunication practices between the respondent and another person. Respondents rate each statement on a 4-point scale from "Never" to "Often." We adapted an 8-item version of the MSQ to omit items that were specific to the supervisory context (e.g., items related to managing challenges at work) and made minor language changes to refer to friendship.

- (1) We talk directly about the interactions in our friendship.
- (2) We monitor what is working/ not working between us in friendship.
- (3) My friend checks in with me about my feelings about what is happening between us during our interaction.
- (4) When I am worried that my friend may be feeling something negative towards me (e.g. frustration), we talk about this together directly.
- (5) When I feel uncomfortable or upset about something that happens in our interaction, my friend and I discuss this openly.

- (6) We speak about how comfortable we feel to discuss things openly with one another in our friendship.
- (7) We speak about things that may have previously been censored, concealed or unsaid in our relationship.
- (8) When my friend and I have a difference of opinion, we discuss this together openly.

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